

PV–PWR–4 power board

Rev A2 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PV–P100/110: 4 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

01 PC connector, command buffers, status

Interfaces PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points

File: sheets/01_pc.kicad_sch

02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/02_ph1_power.kicad_sch

03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03_ph1_gda.kicad_sch

04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04_ph1_gdb.kicad_sch

05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/05_ph2_power.kicad_sch

06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06_ph2_gda.kicad_sch

07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07_ph2_gdb.kicad_sch

08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/08_ph3_power.kicad_sch

09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09_ph3_gda.kicad_sch

10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10_ph3_gdb.kicad_sch

11 Phase 4 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/11_ph4_power.kicad_sch

12 Phase 4 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/12_ph4_gda.kicad_sch

13 Phase 4 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/13_ph4_gdb.kicad_sch

14 Port A power path (pv)

Varistor Y + monitor

File: sheets/14_leanA_power.kicad_sch

15 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/15_leanA_ctrl.kicad_sch

16 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/16_leanA_drive.kicad_sch

16 Port B power path (battery)

CALCULATED: not measured. design_check() asserts every number on each build (outputs: PV–PWR–4_schematic_checks.txt), drafted @ few days from design, 06/09/2026. Domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651ASC–Q1, T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX–T1 reinforced winding -> J_SELV). Coat to PD1. Shared heatsink: Tj 110 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 23.7 W (economiser) <= aux live rating 24.0 W; module aux 80 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

References: PV–PWR–4_schematic_checks.txt, drafted @ few days from design, 06/09/2026. polarity and dV interlocks

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/18_leanB_drive.kicad_sch

18 Terminals, EMI filter, insulation test

DC terminals, nanocrystalline CM ring per port, X / Y capacitors, board entries (press–fit), PE links, heatsink, insulation monitor (port.lean_imd)

File: sheets/18_terminals_imd.kicad_sch

19 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/19_live5v.kicad_sch

20 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–: 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uF / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/20_aux75_input.kicad_sch

21 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 35–80 ms

File: sheets/21_aux75_start.kicad_sch

22 AUX75 PWM, SIC, clamp, AUX–T1

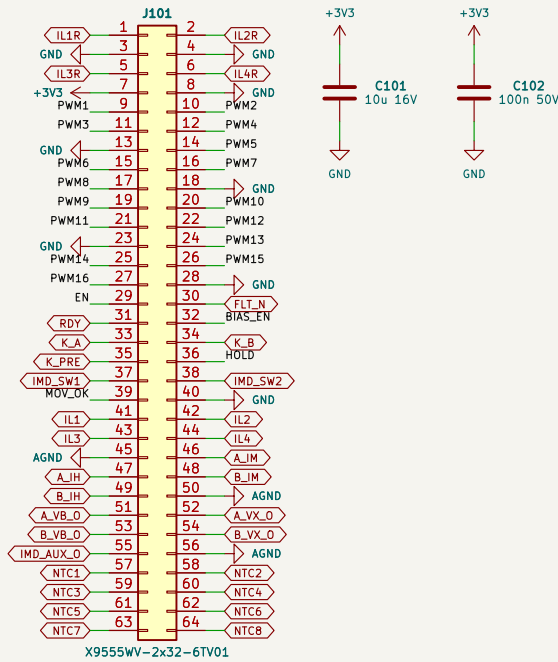
UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 48:8:9; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/22_aux75_pwm.kicad_sch

Not bench–validated. Values read from sim/out/*.json; checks: design_check()
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

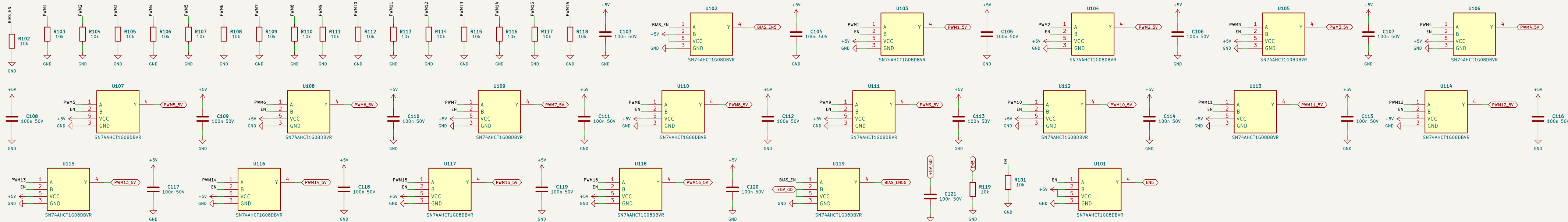
Sheets: 1
File: F
Title
Size: 6
KiCad

PC CONNECTOR (INTERFACES.PC), LIVE



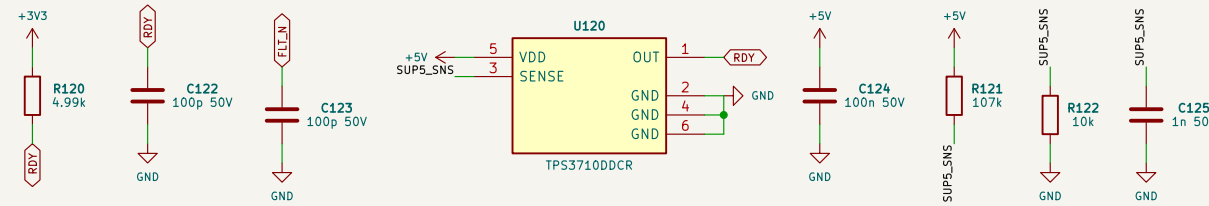
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2: the former +24V / +5V pins carry the sensor references (ILR-IL4R)). Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



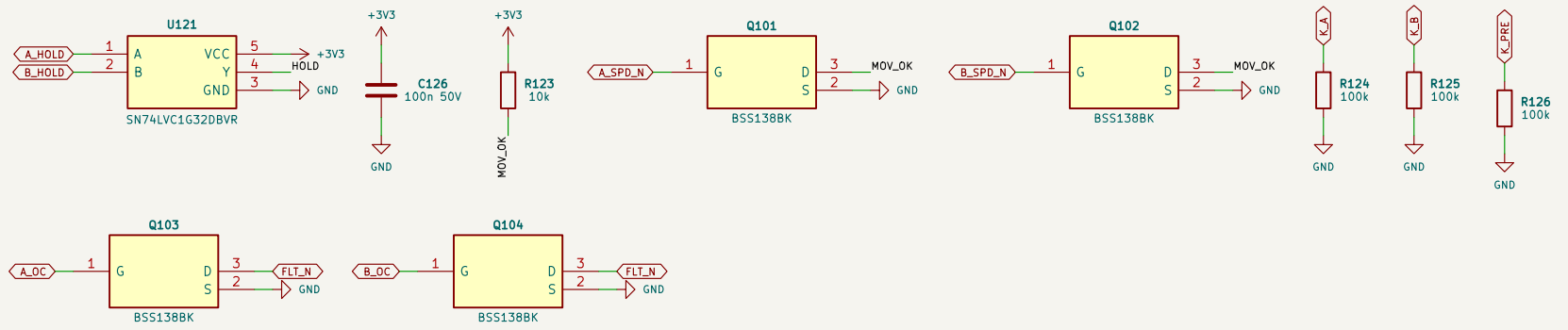
NSI6651 inputs are specified at VCC1 = 5 V only (VIN1 <= 3.3 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



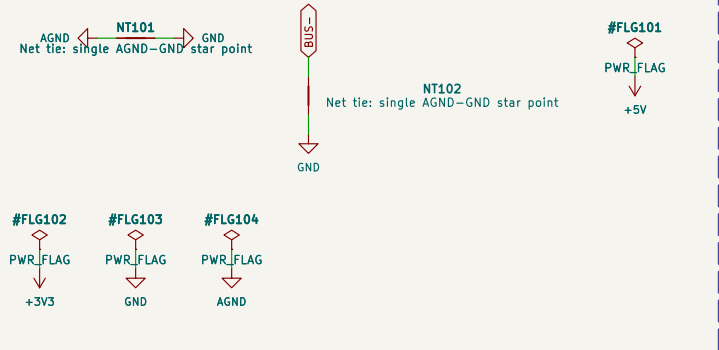
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC (pull-up on the control board, +5V supervisor TPS3710 107V/20n: RDY low below 4.52–4.63 V (AHCT VCC >= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



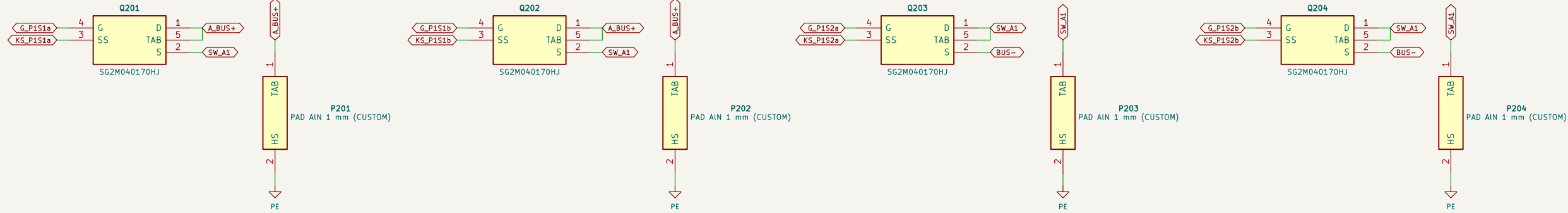
HOLD = A_HOLD OR B_HOLD ([I_port] >= 0.97–1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([I] >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



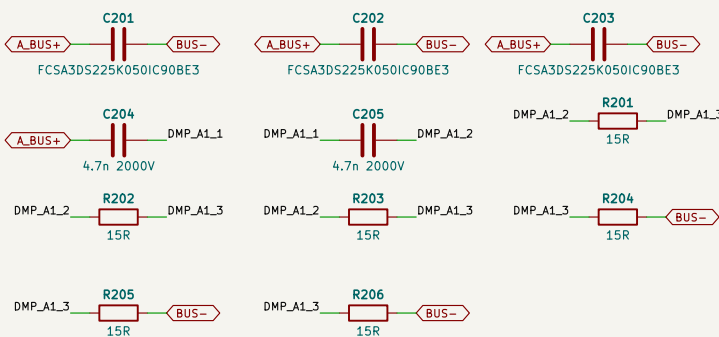
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-). Layouts gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



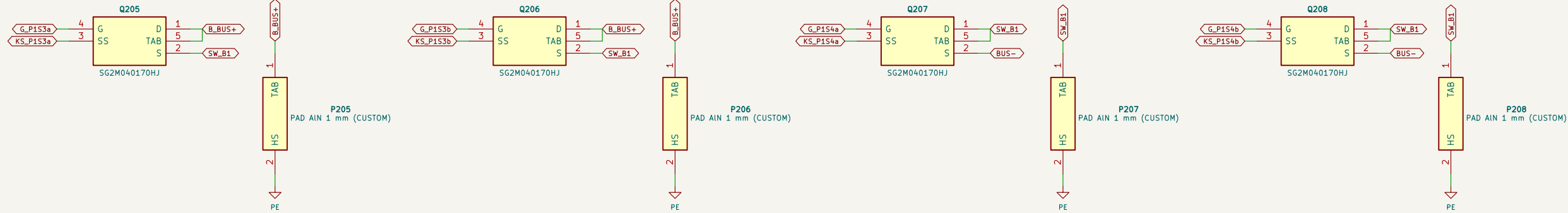
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



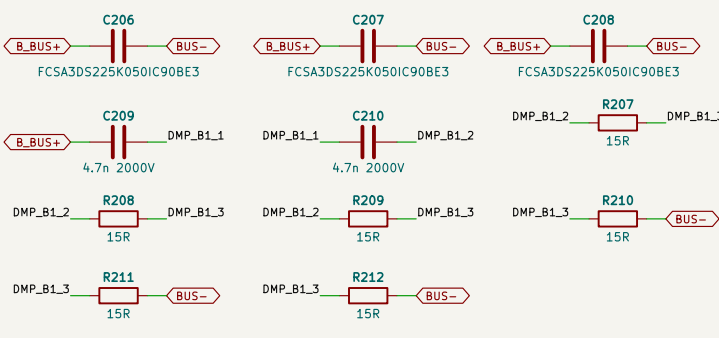
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



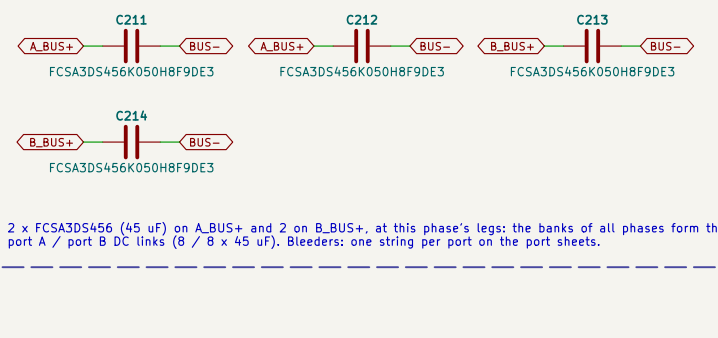
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



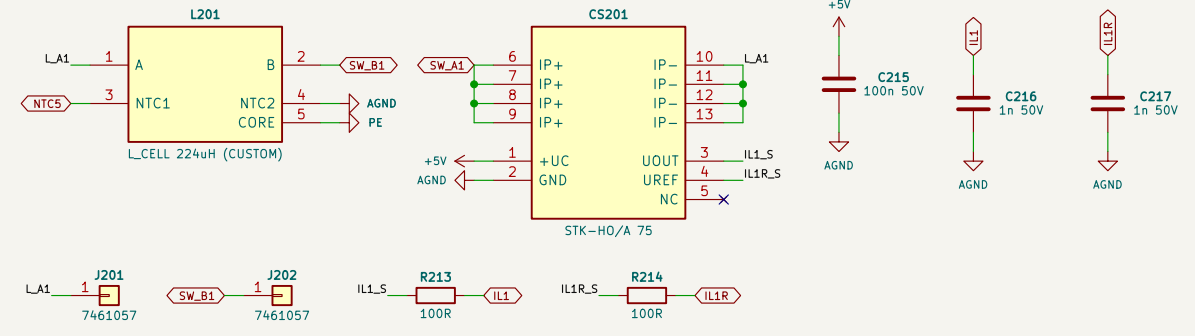
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



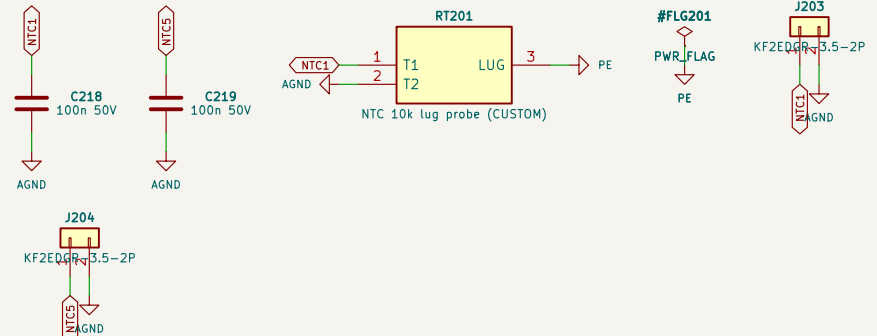
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



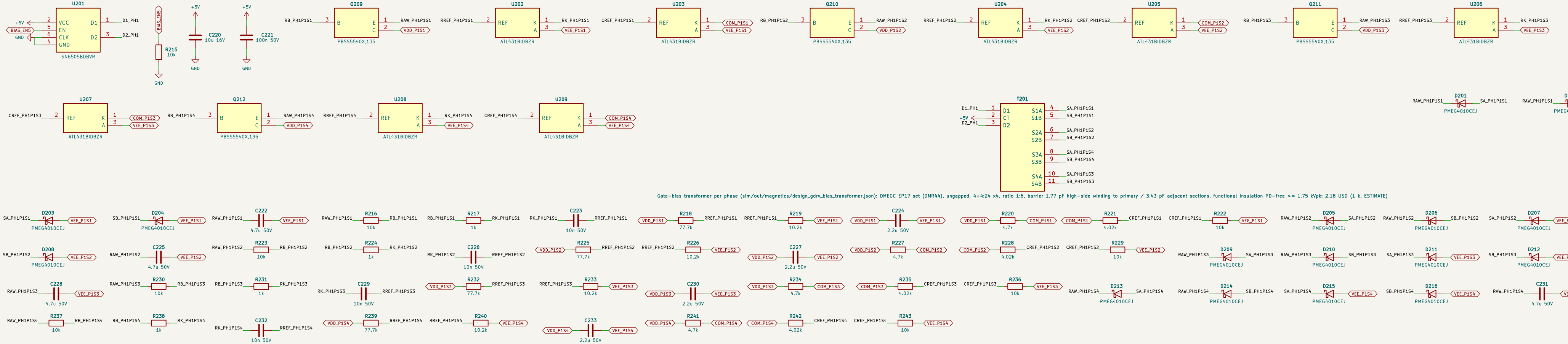
Lead A from SW.A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW.A -> SW.B) to stud
L.A1: lead B on stud SW.B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R = 1 nF at the source, IL1R = the sensor's own (ref pin 6), 100R + 1 nF (PVR-CTL) references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC5 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

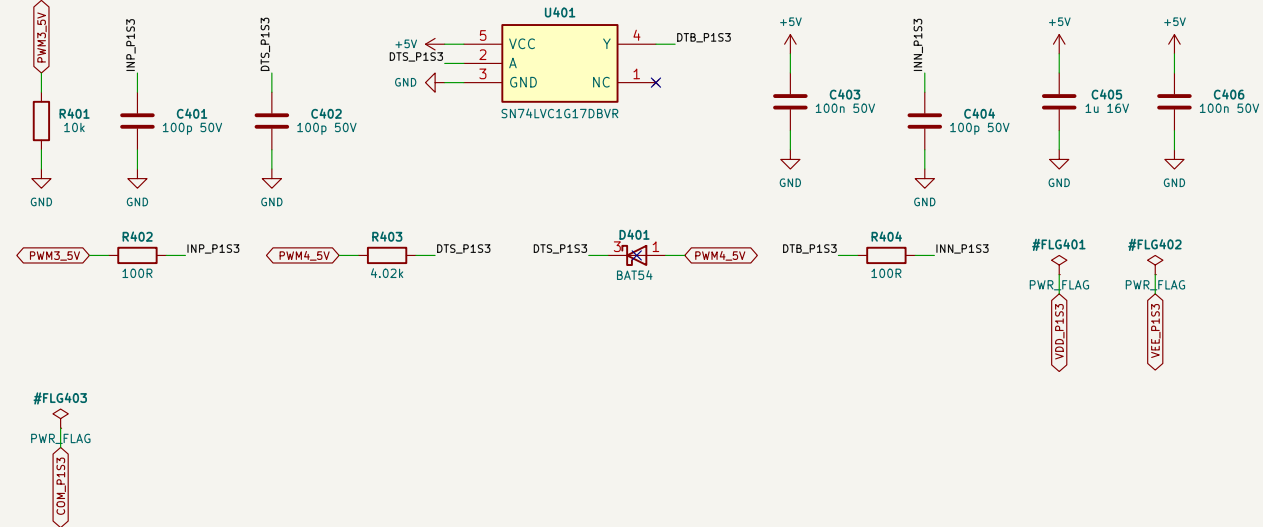
PH1: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMCC EP17 set (DMR44), ungapped, 4+4:2k v4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 UOV (1 k, ESTIMATE)

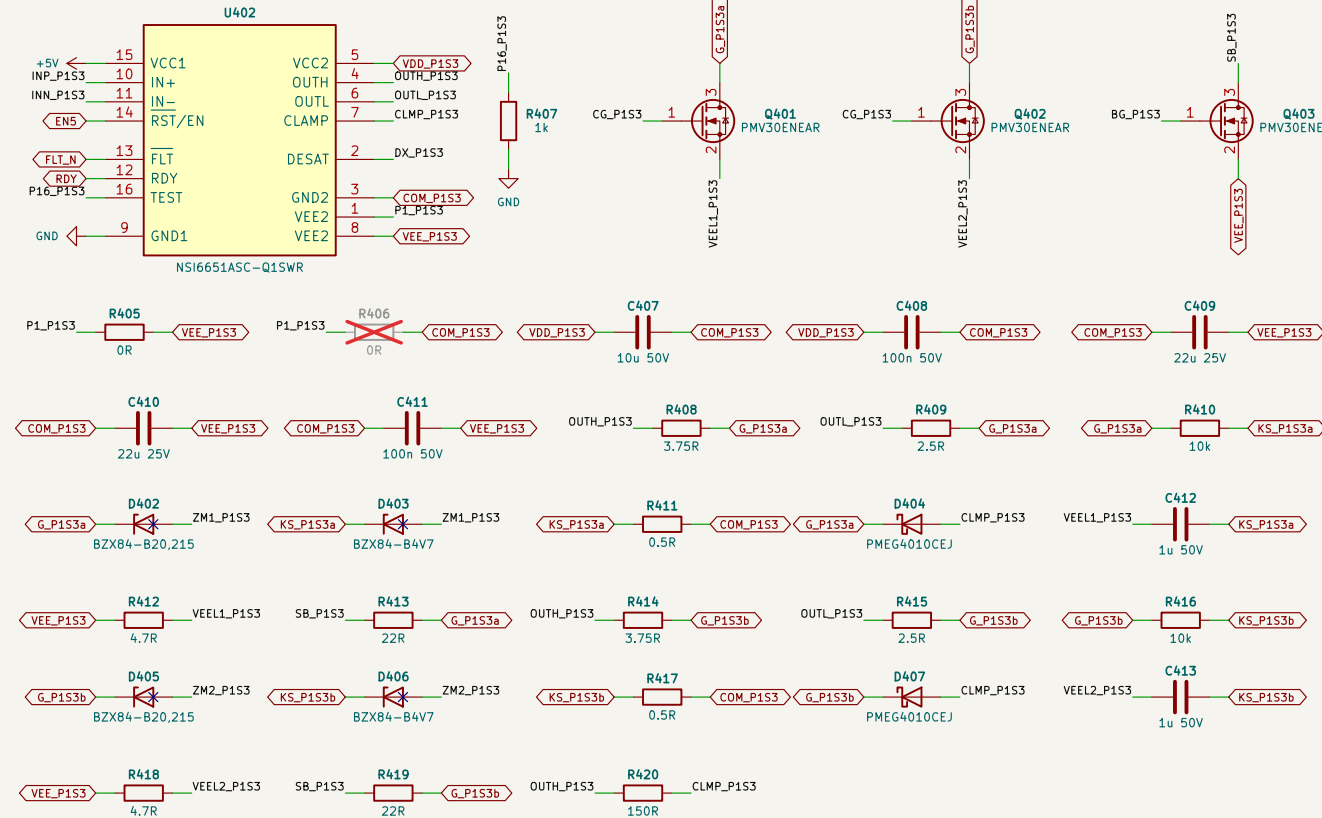
SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T.BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S3: LOGIC SIDE



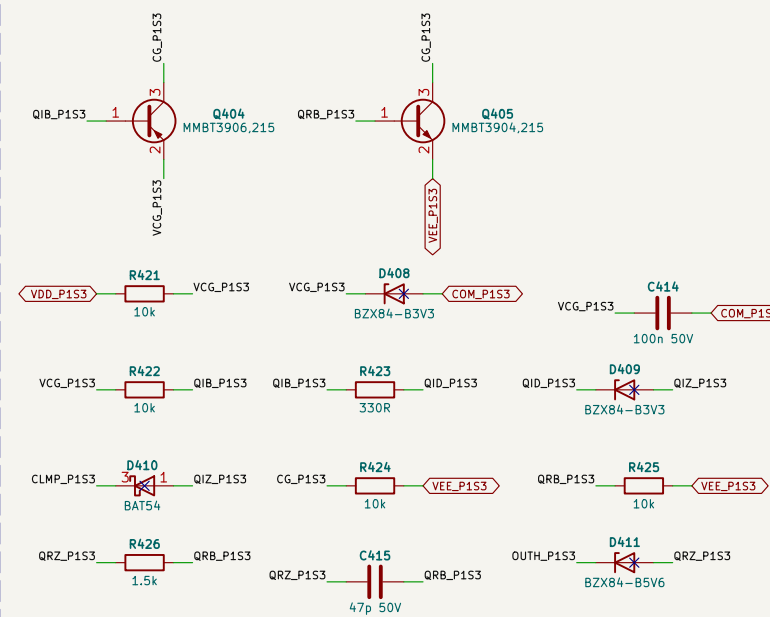
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



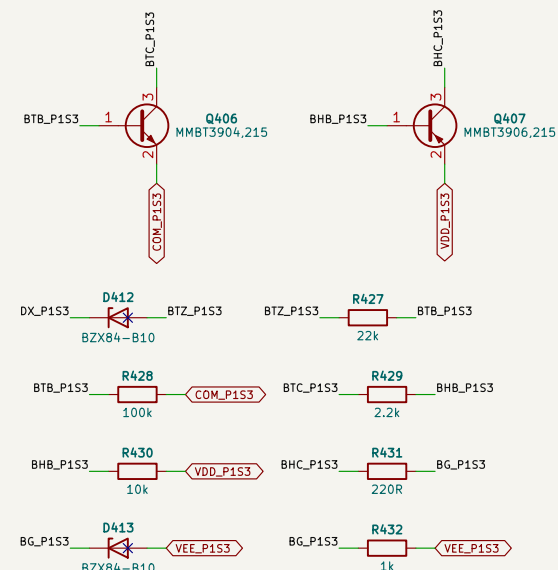
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



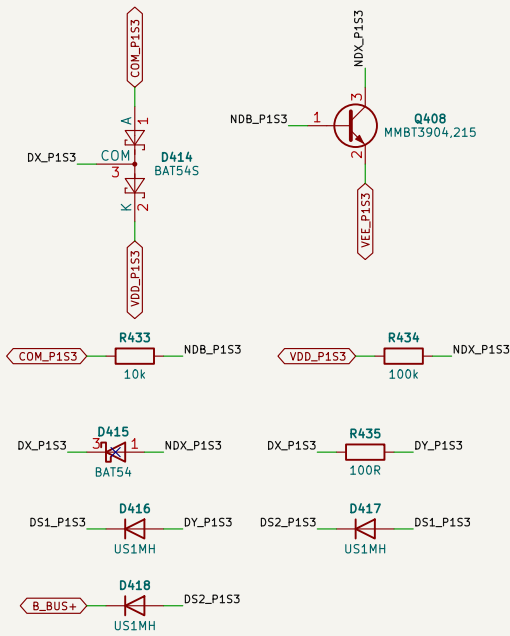
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



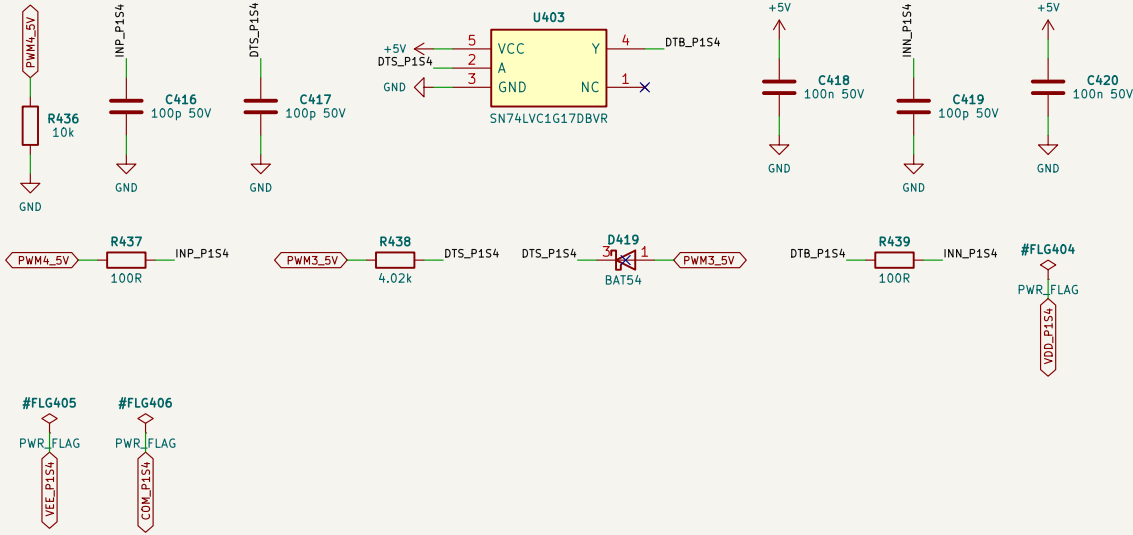
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P1S3: DESAT



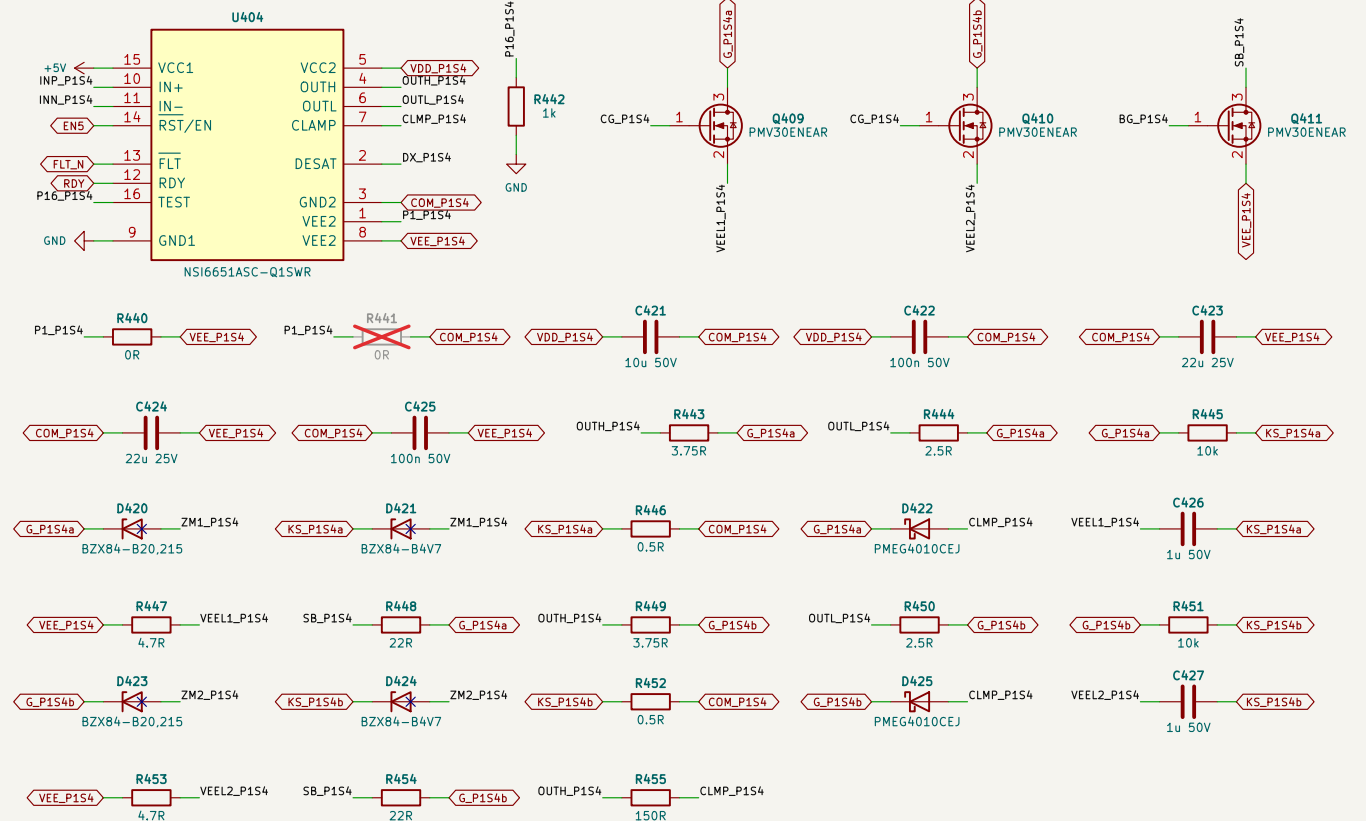
DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

P1S4: LOGIC SIDE



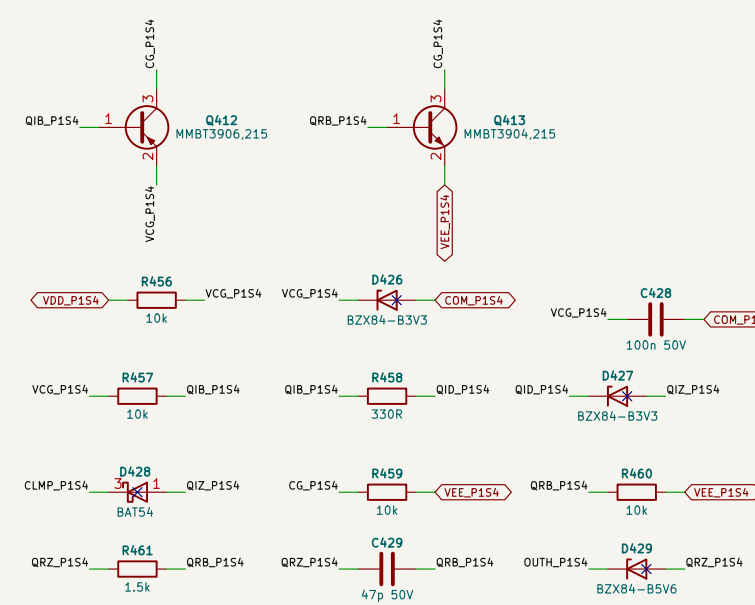
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



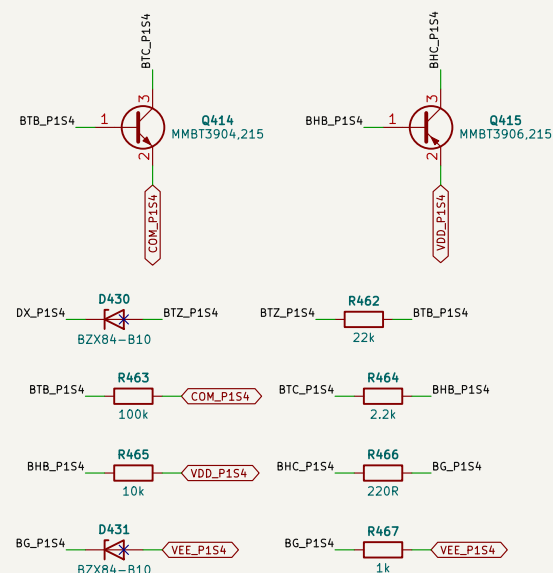
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH

P1S4: MILLER CLAMP CONTROL



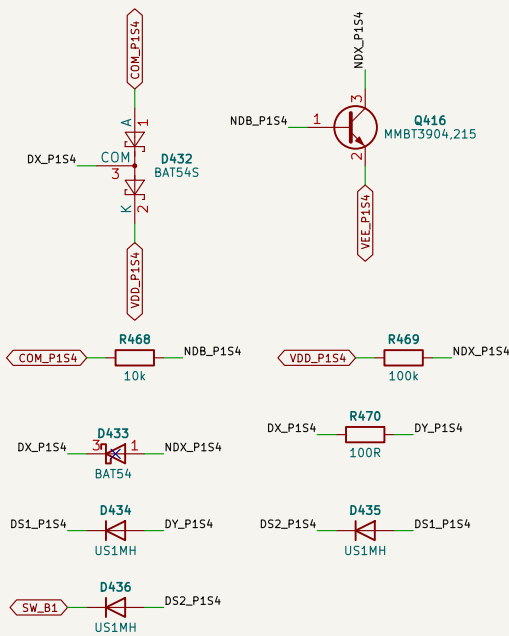
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3–10.9 V (above the 9.8 V maximum trip) → MMBT3904 → MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P1S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 5
Schematics authored by Chinmoy Bhuyan

Sheet: /04 Phase 1 leg B gate drive (S3, S4)/
File: 04_ph1_gdb.kicad_sch

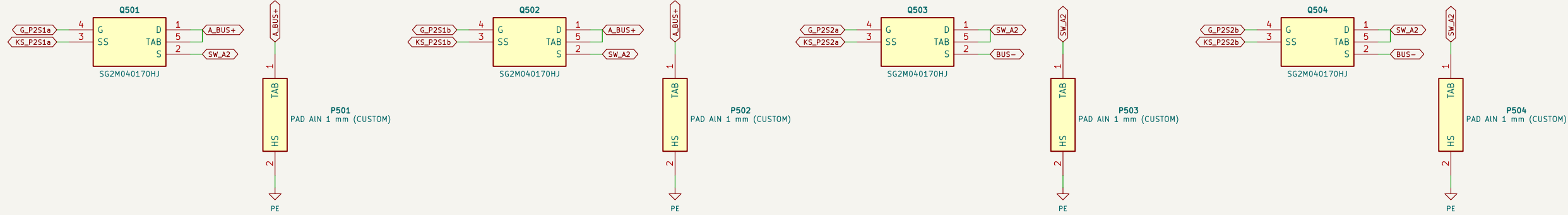
Title: PV-PWR-4 power board – Phase 1 leg B gate drive (S3, S4)

Size: A2	Date: 2026-10-05
----------	------------------

KiCad E.D.A. 10.0.6 Id: 5/26

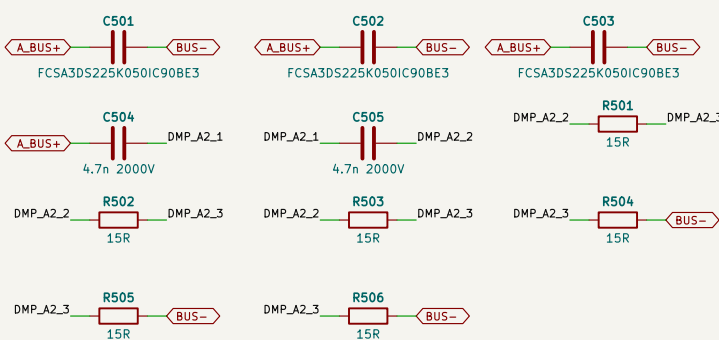
Id: 5/26

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



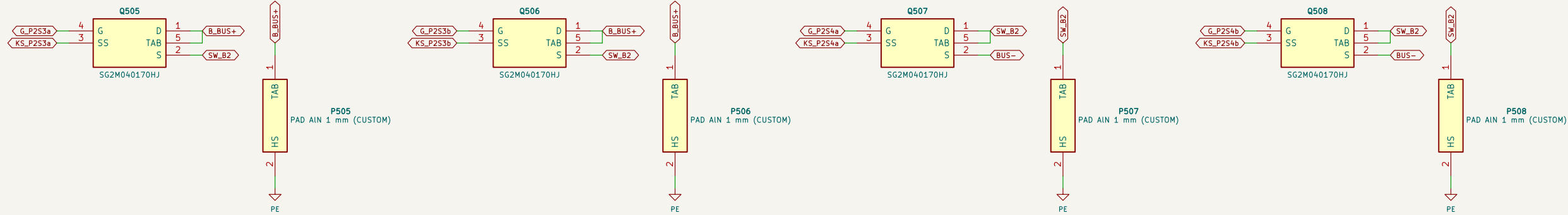
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



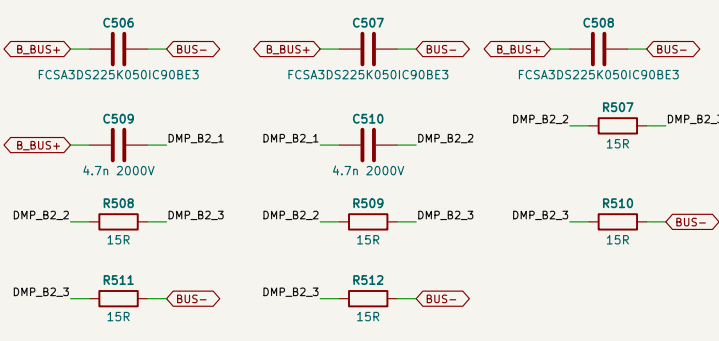
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



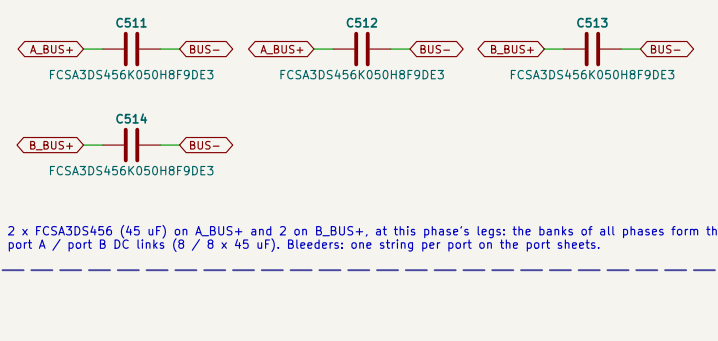
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



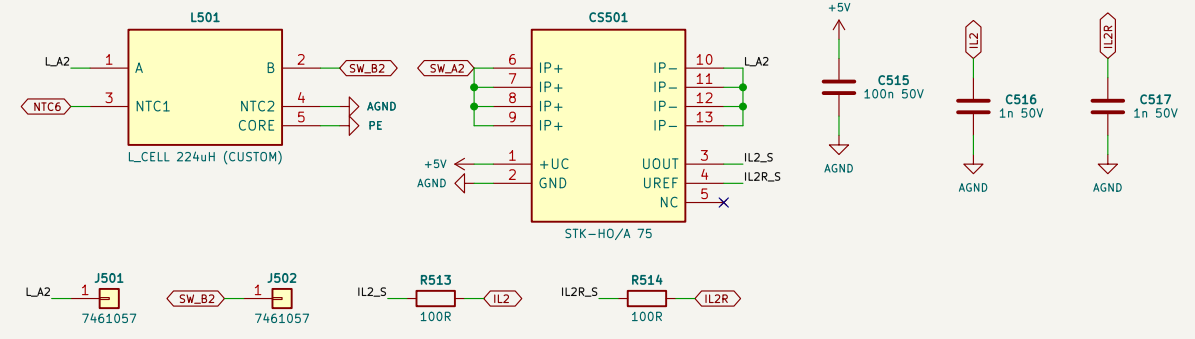
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



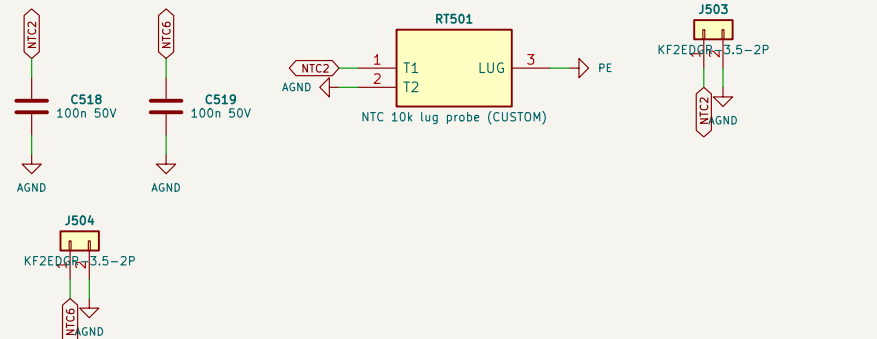
2 x FCSA3DS456 (45 uf) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (8 / 8 x 45 uf). Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



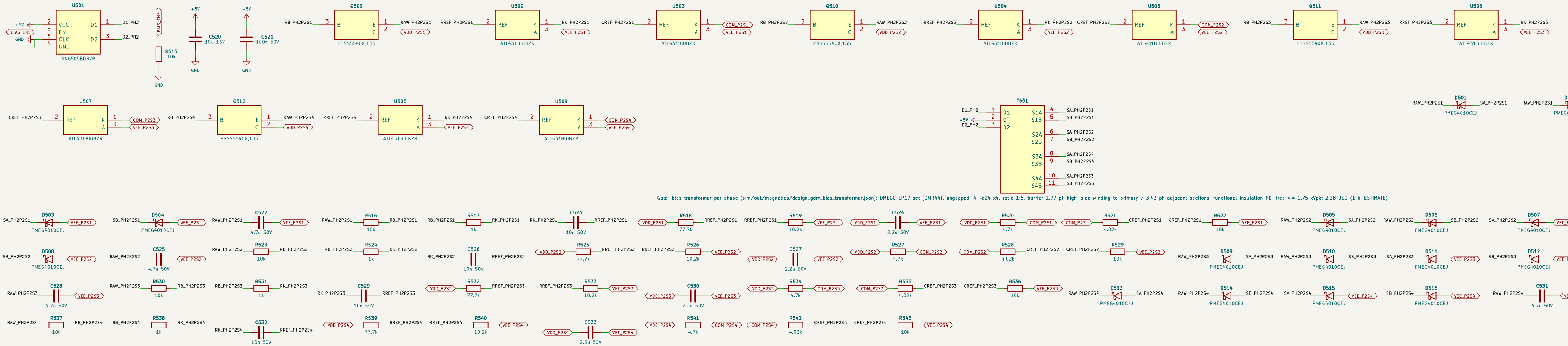
Lead A from SW_A2 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud
LA2: lead B on stud SW_B2, IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R = 1 nF at the source, IL2R = the sensor's own (ref pin 6), 100R = 1 nF (PVR-CTL references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



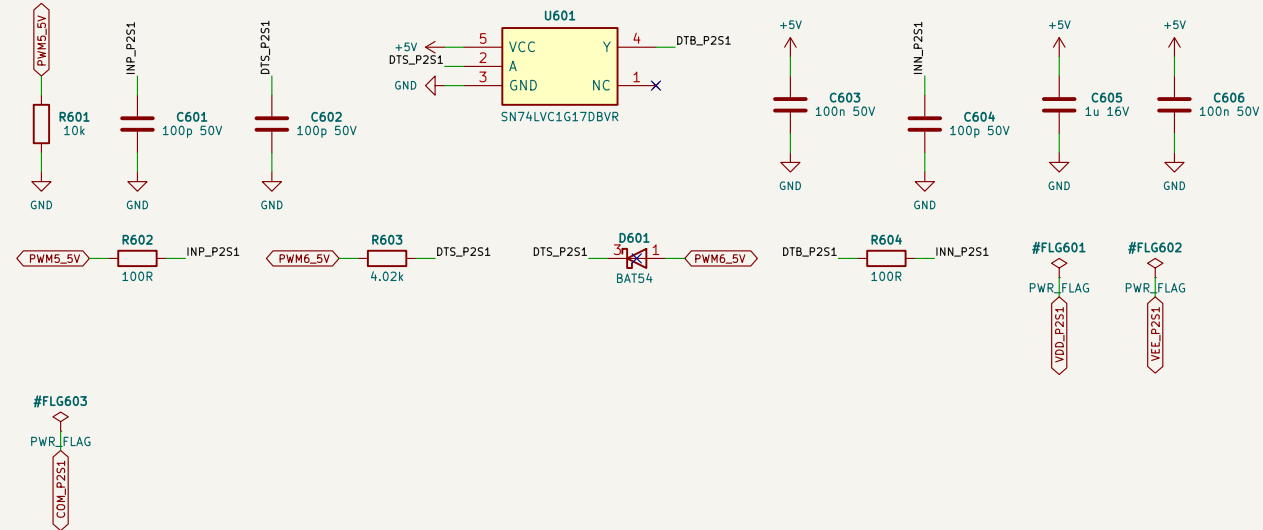
NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC6 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



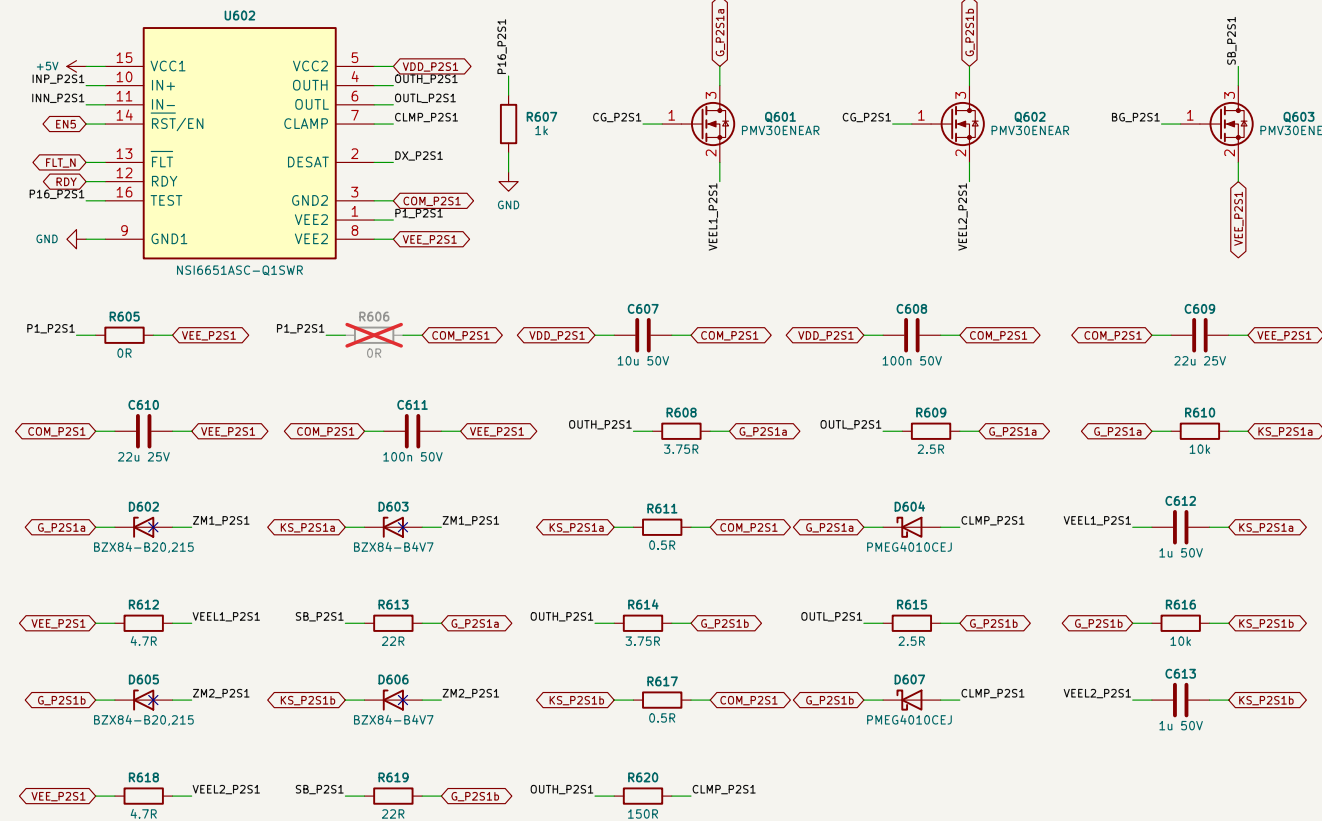
SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431, COM split.

P2S1: LOGIC SIDE



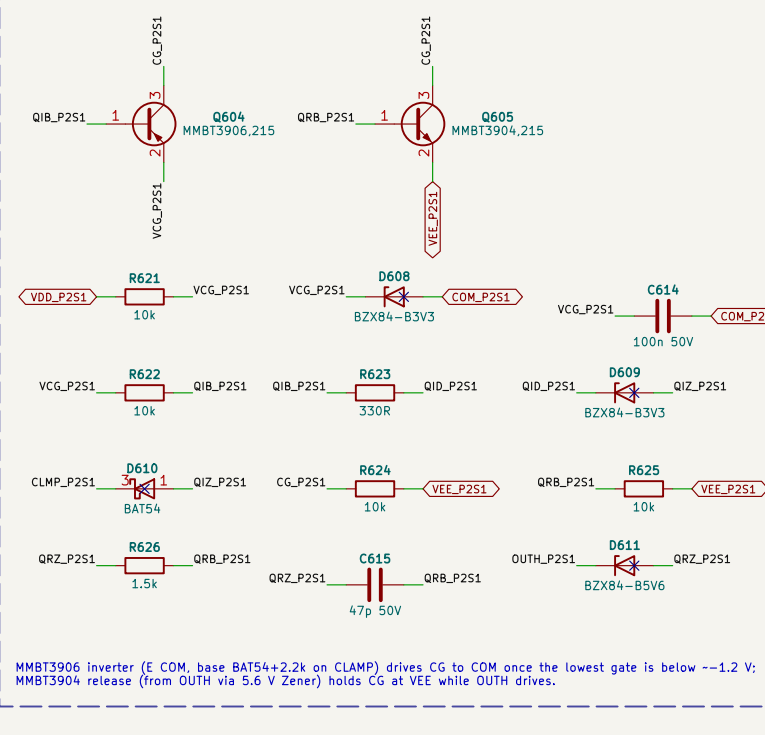
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



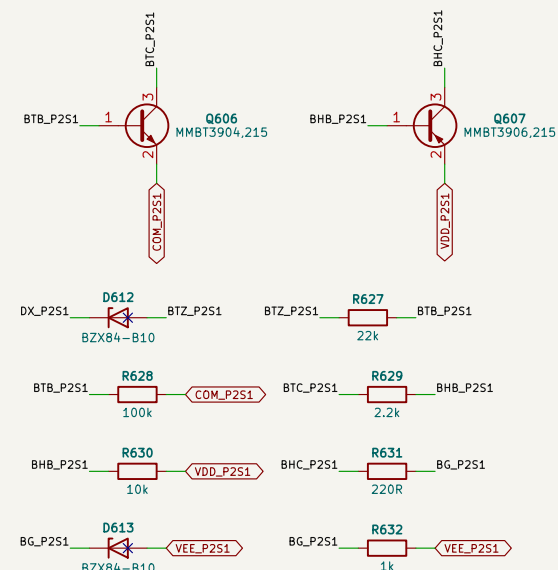
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL



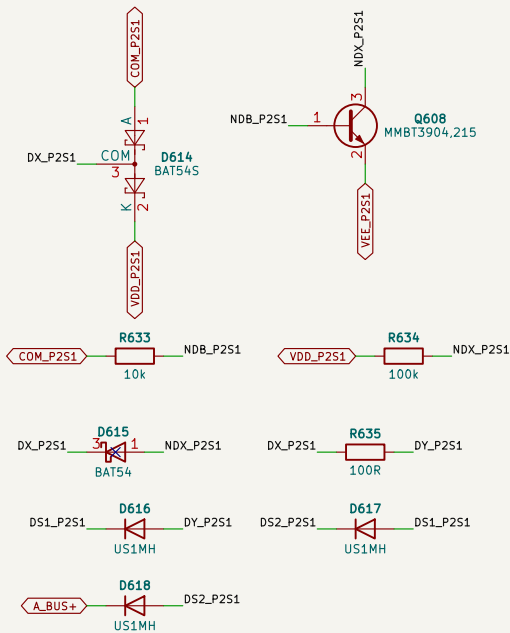
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER



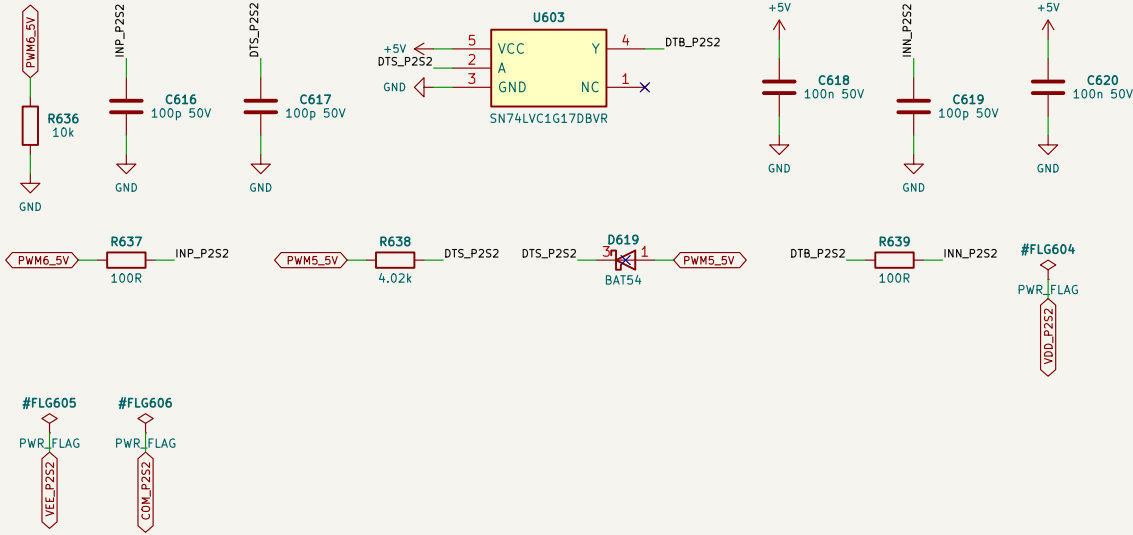
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V
per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P2S1: DESAT



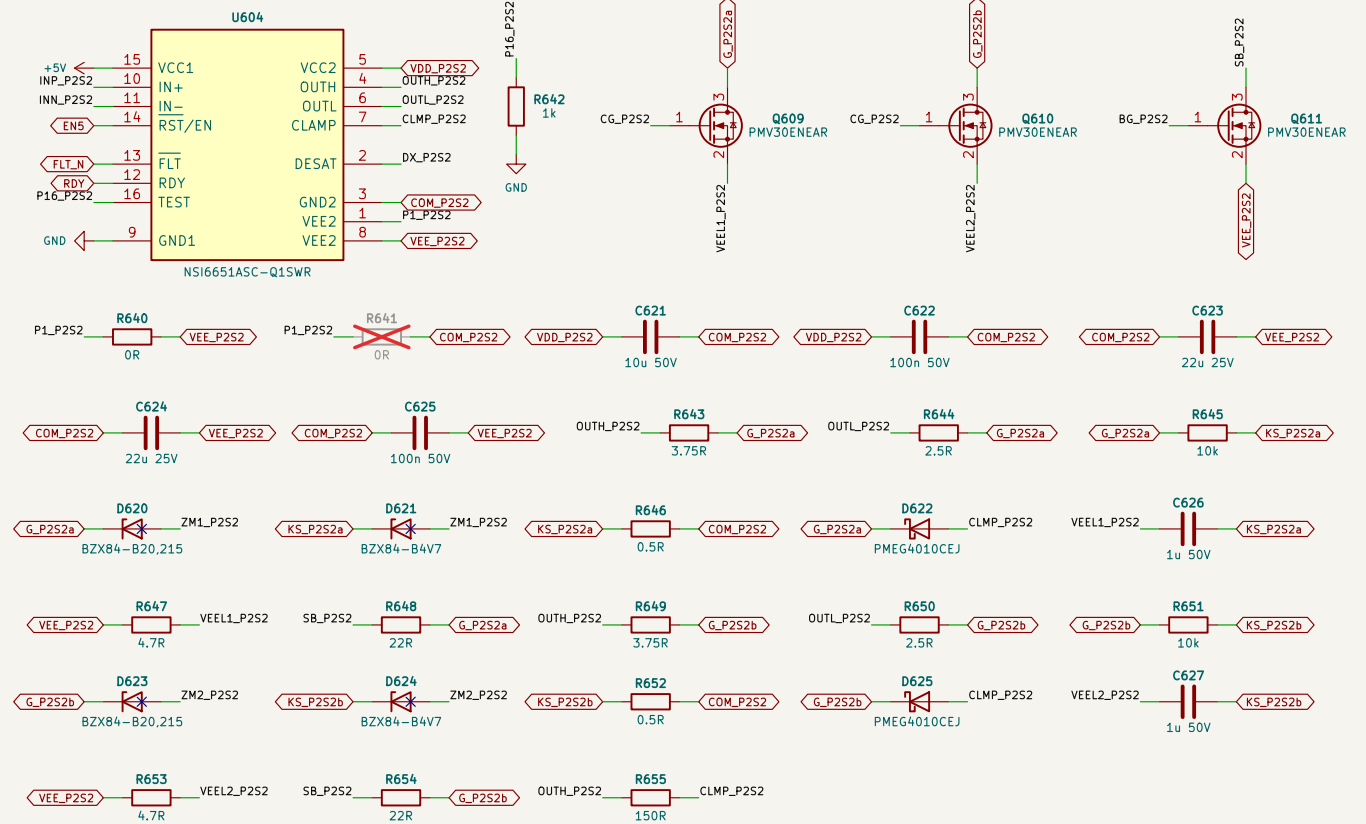
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S2: LOGIC SIDE



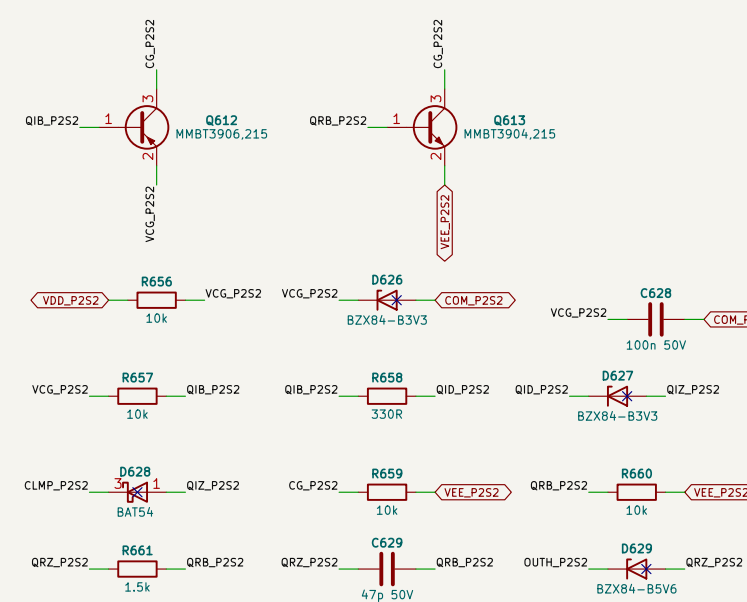
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S2: DRIVER AND 2 GATES



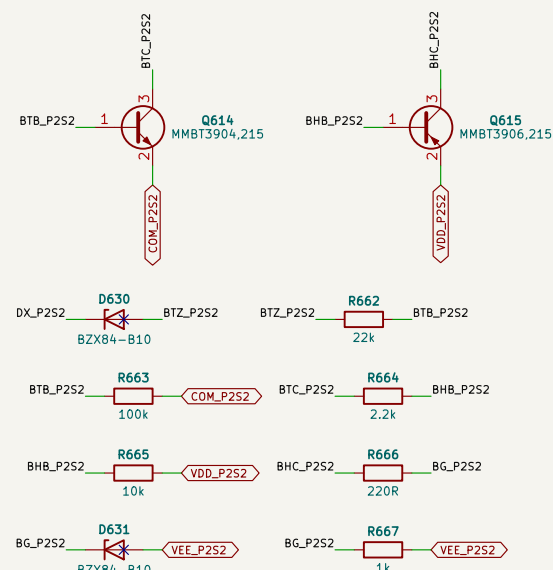
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH

P2S2: MILLER CLAMP CONTROL



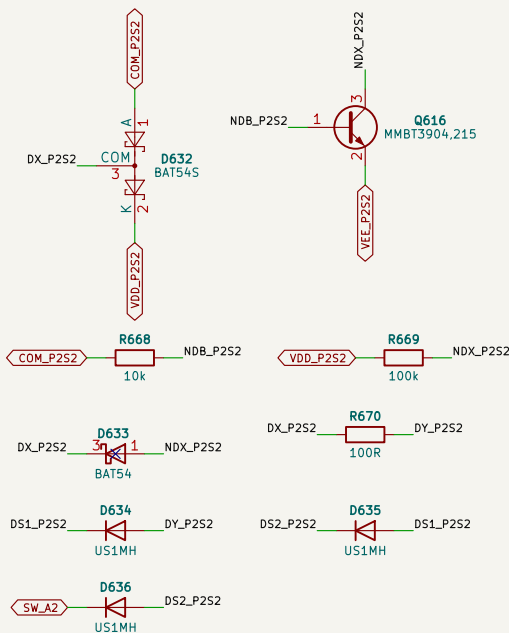
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AO3400A + 22k pull the gate to VEE; BG holds -3 us (1k + FET gates).

P2S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 7
Schematics authored by Chinmoy Bhuyan

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/
File: 06_ph2_gda.kicad_sch

Title: PV-PWR-4 power board – Phase 2 leg A gate drive (S1, S2)

Size: A2	Date: 2026-10-05	Rev: A2
----------	------------------	---------

KiCad E.D.A. 10.0.6		Id: 7/26
10	11	

[illegible]

The schematic diagram illustrates the NS16651ASC-Q15WR driver circuit. At the top left, the U702 driver IC is shown with its pin connections: VCC1 (5V), VCC2 (VDD_P253), IN+ (IN_P253), IN- (IN_P253), OUTL (OUT_P253), CLAMP (CLMP_P253), DESAT (DX_P253), RDY (RDY_P253), TEST (TEST_P253), and GND1 (VEE_P253). The driver IC is connected to a 5V supply and a 1k resistor (R707) to ground. The output of the driver IC is connected to the gates of four PMOSFETs (Q701, Q702, Q703, and Q704) through a 1k resistor (R707). The PMOSFETs are connected to a 5V supply and have their sources connected to ground. The gates of the PMOSFETs are controlled by the driver IC. The circuit includes various passive components such as resistors (R705, R706, R707, R708, R709, R710, R711, R712, R713, R714, R715, R716, R717, R718, R719, R720) and capacitors (C707, C708, C709, C710, C711, C712, C713) for timing and protection. The circuit is designed to drive a load, with various components like resistors and capacitors used for timing and protection.

[illegible]

Figure 1: Schematic representation of the genetic constructs. The figure shows two main constructs, Q706 and Q707, and their derivatives. Q706 (MHBT3904.215) contains BTB_P253, Q706, and COM_P253. Q707 (MHBT3906.215) contains BHB_P253, Q707, and YDB_P253. Derivatives include: 1) BTB_P253 and BTZ_P253 with a 712k deletion (BZM4-010). 2) BTZ_P253 and BTB_P253 with a 22k deletion. 3) BTB_P253 and BTZ_P253 with a 100k deletion (R728). 4) BHB_P253 and BHC_P253 with a 10k deletion (R730). 5) BHB_P253 and B6_P253 with a 220k deletion (R731). 6) B6_P253 and B6_P253 with a 173k deletion (R732).

.....

[illegible]

[illegible]

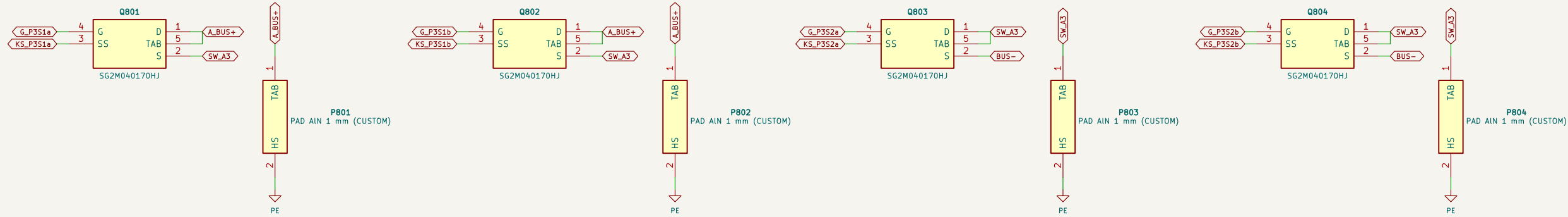
[illegible]

Figure 1: Schematic representation of the genetic organization of the human D732B15 region. The diagram shows the arrangement of genes and exons (numbered 1-4) on a chromosome. Genes shown include COM_P25A, D732 BA75A5, VDB_P25A, R768, NDB_P25A, R769, VDB_P25A, DX_P25A, D733 BA75A, ND_X_P25A, D734, US1MH, DY_P25A, D735, US1MH, D52_P25A, D51_P25A, D736, US1MH, D52_P25A, SW_B2, and D716. A scale bar indicates 10k and 100k. A legend indicates that red boxes represent genes and blue boxes represent exons.

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string <= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

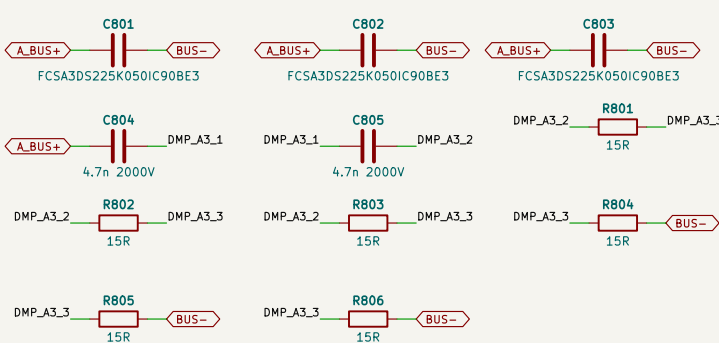
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string <= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



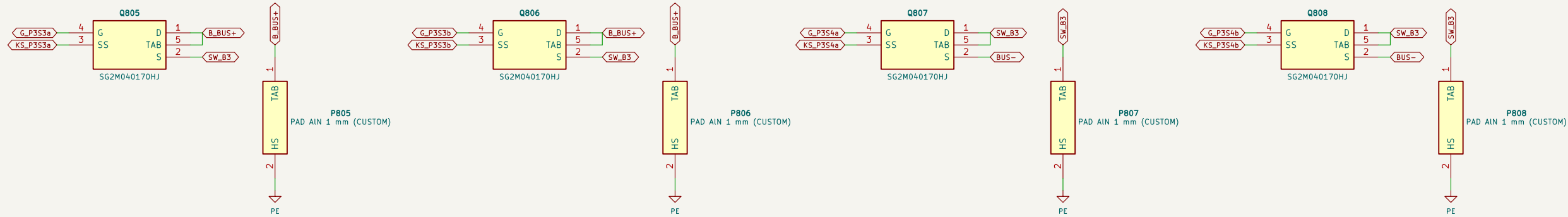
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



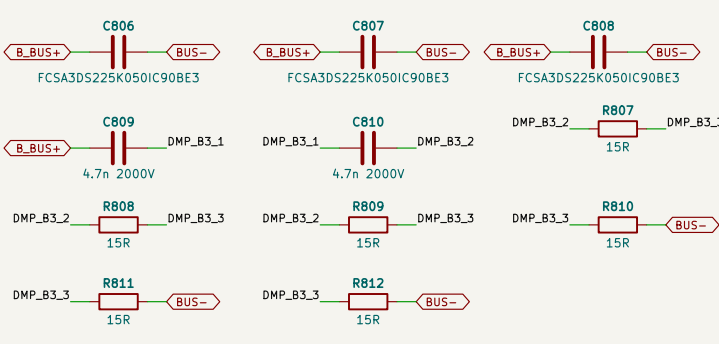
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



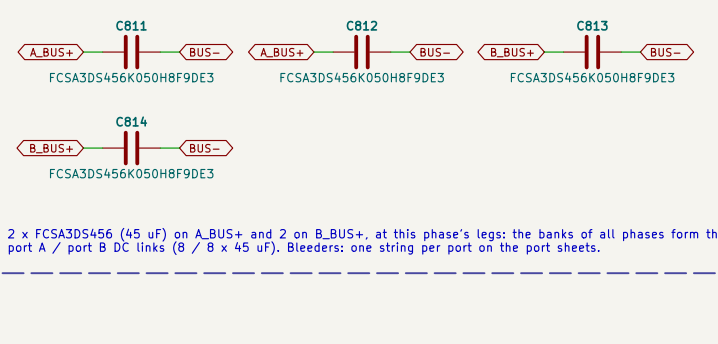
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



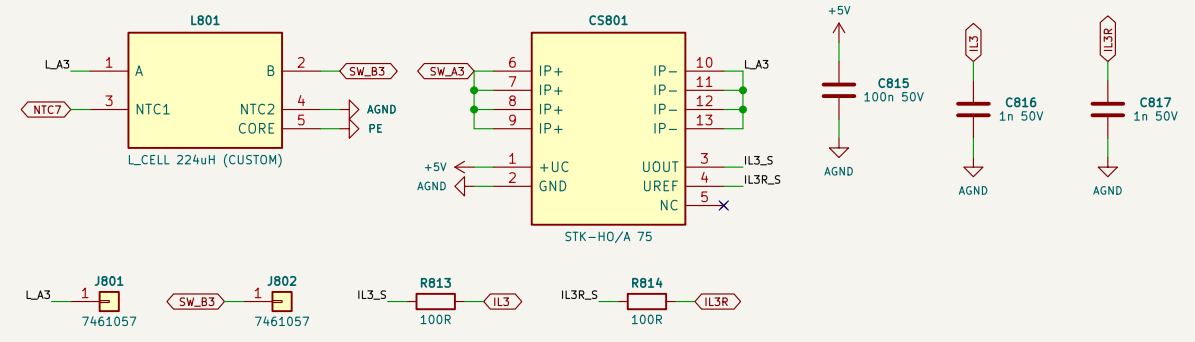
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



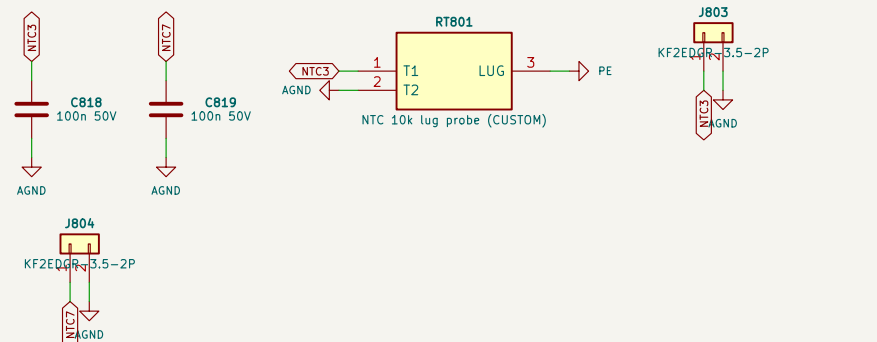
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



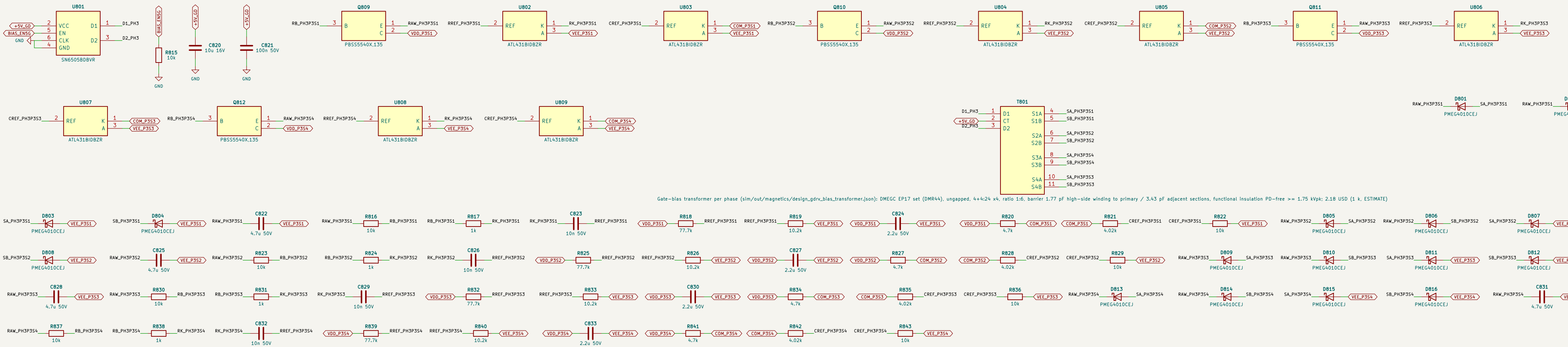
Lead A from SW_A3 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW_A -> SW_B) to stud
LA3: lead B on stud SW_B3, IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-0.5 V over +/-187.5 A;
100R = 1 nF at the source, IL3R = the sensor's own tref (pin 6), 100R + 1 nF (PVC-CTL references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



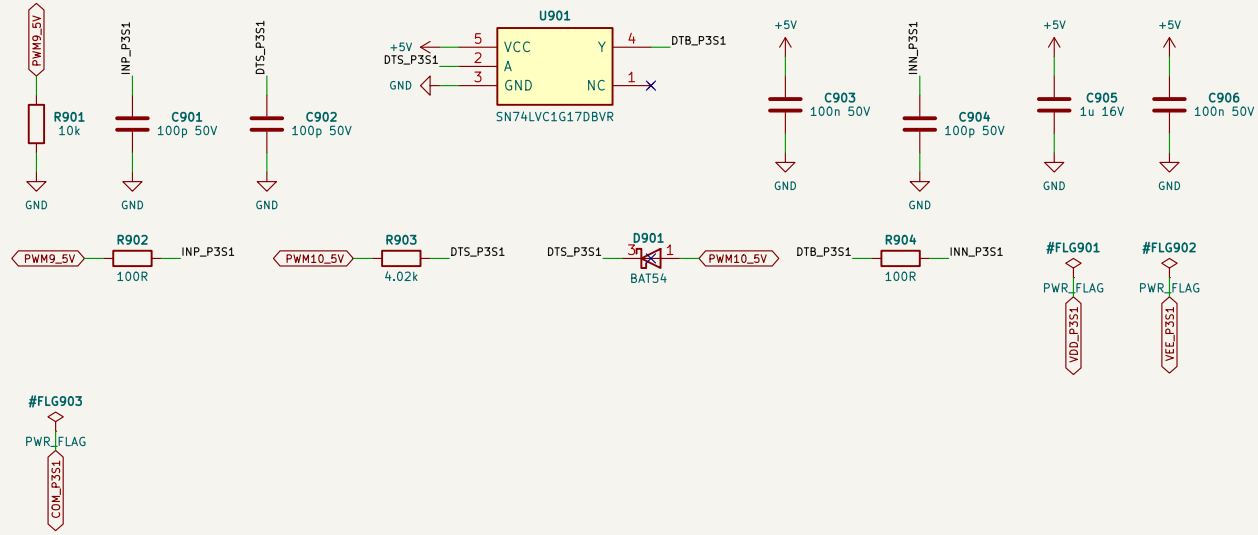
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC7 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



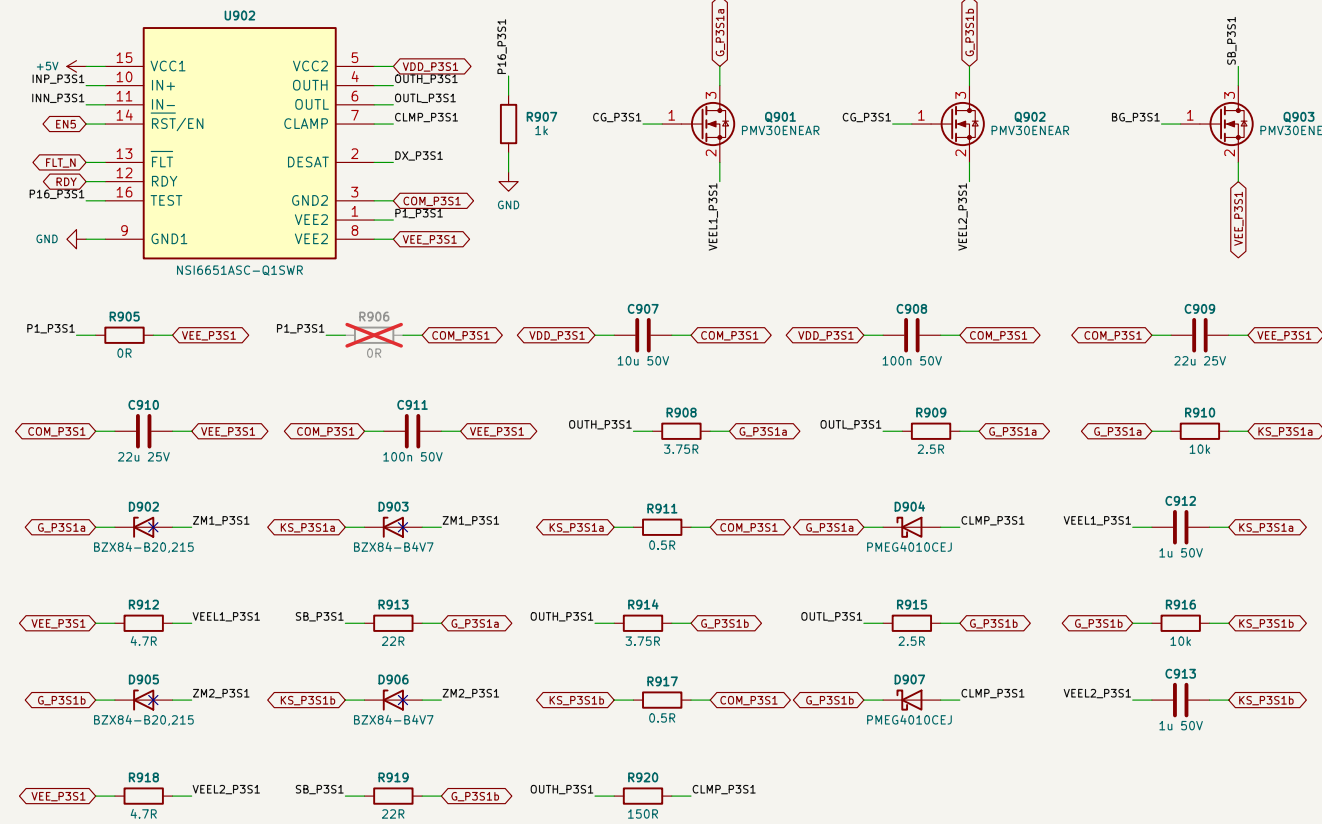
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



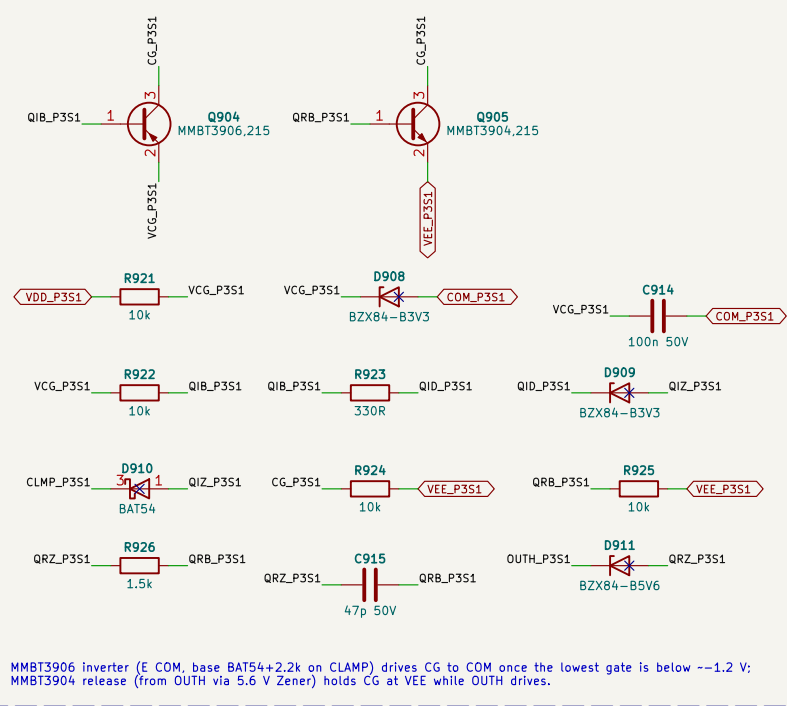
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



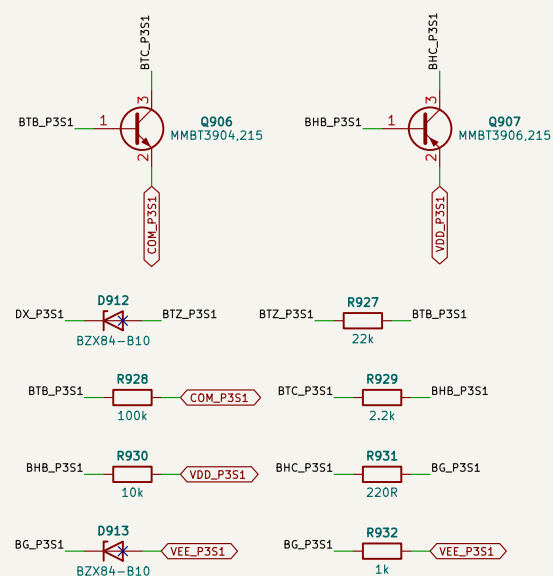
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



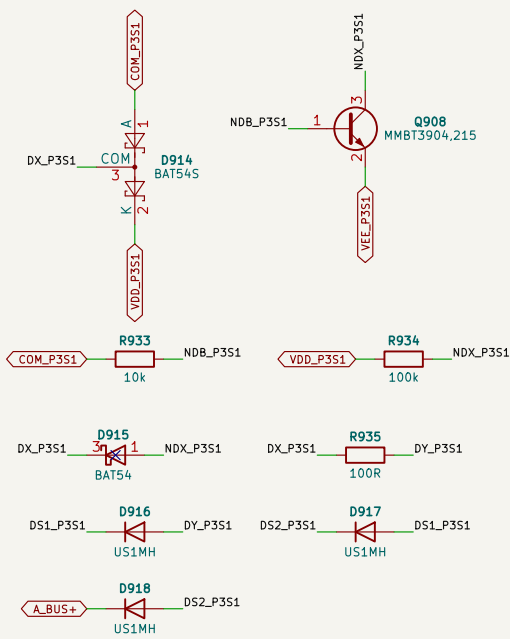
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



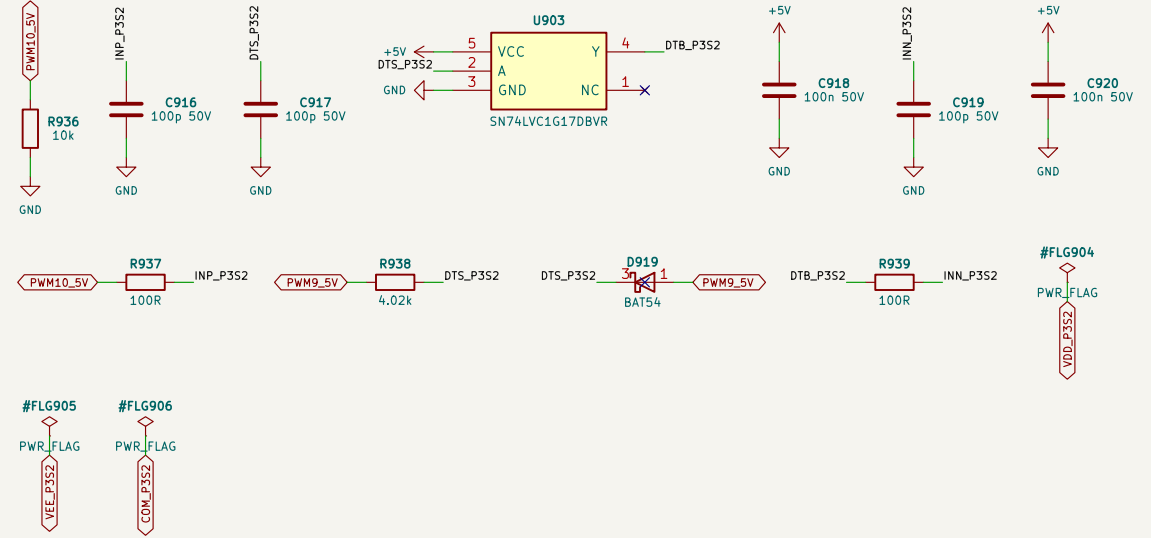
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S1: DESAT



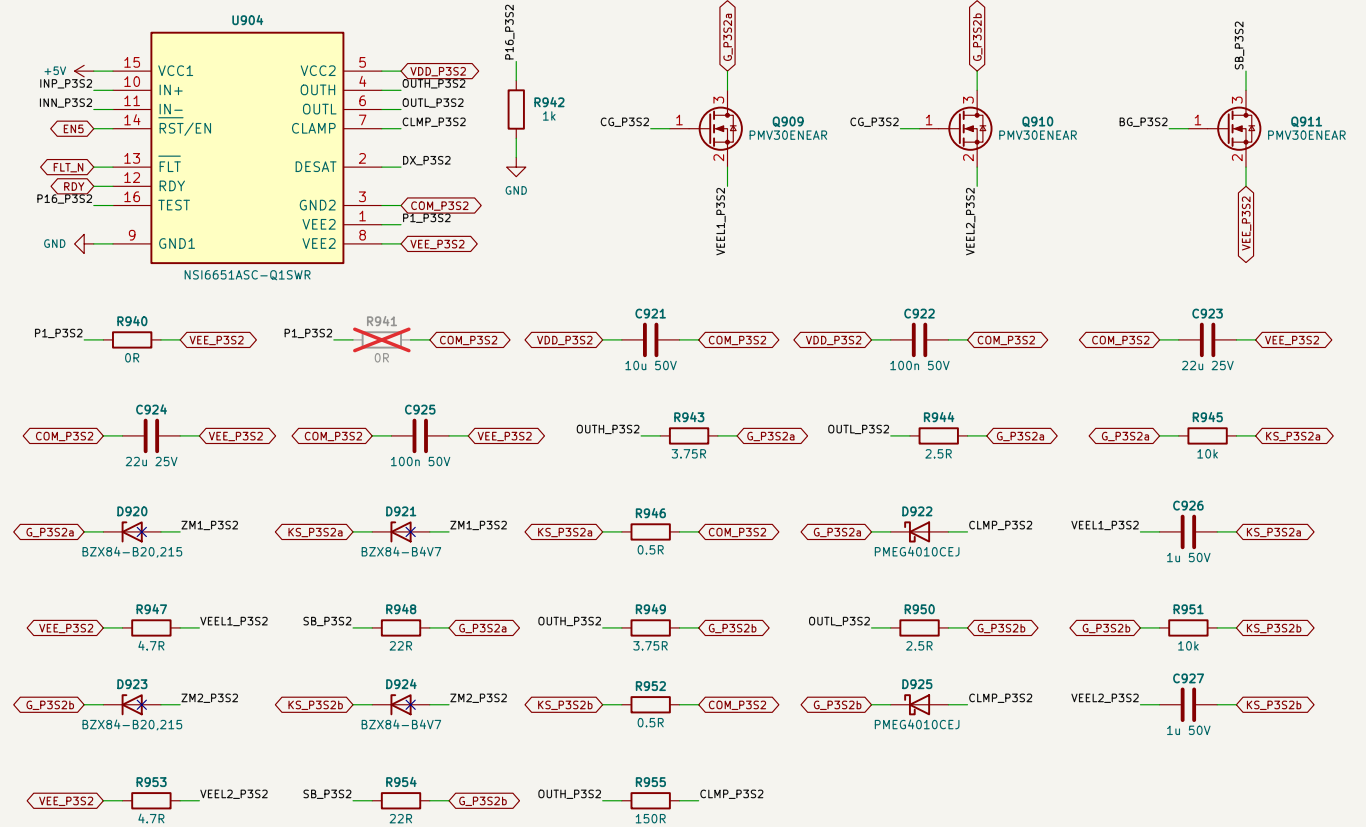
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs - 3 x US1MH - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P3S2: LOGIC SIDE



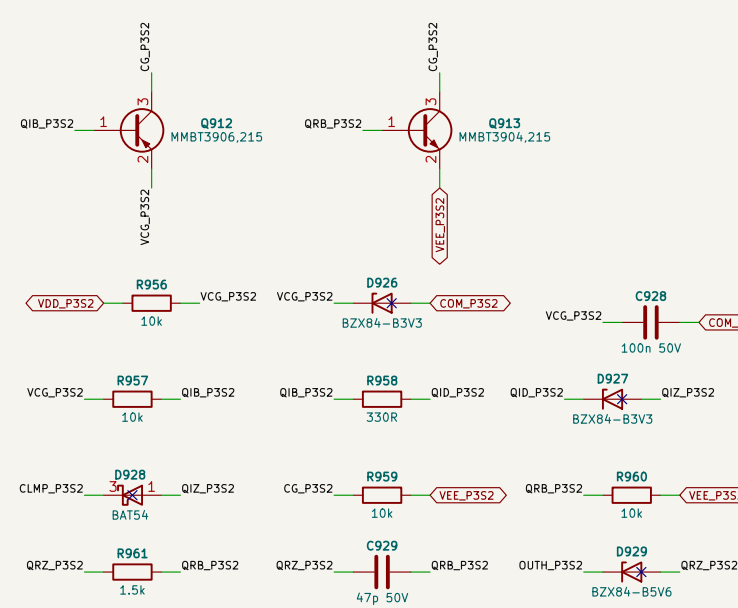
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



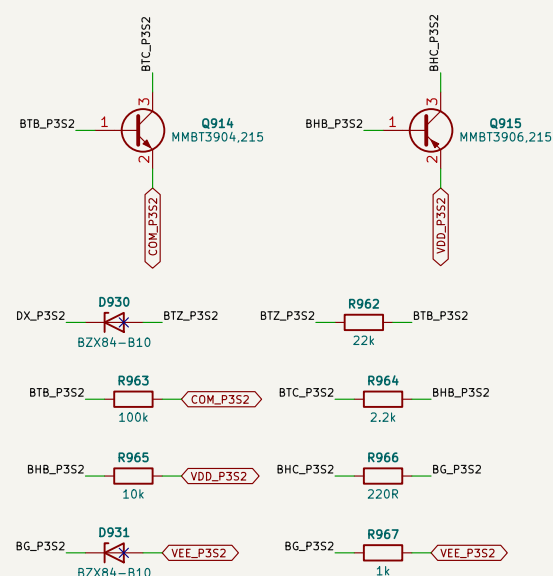
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



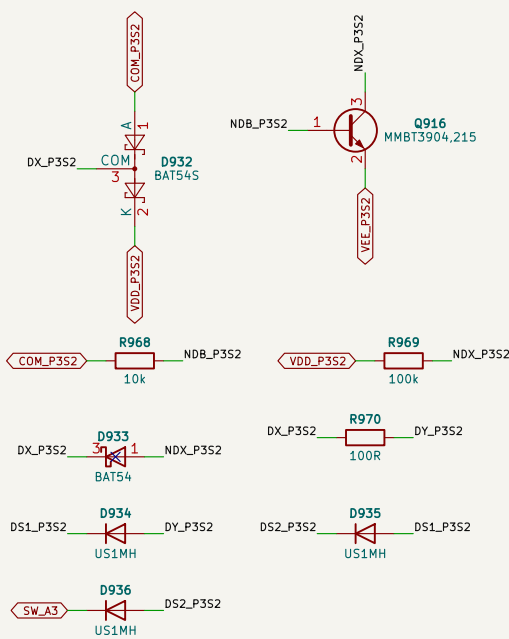
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S2: DESAT

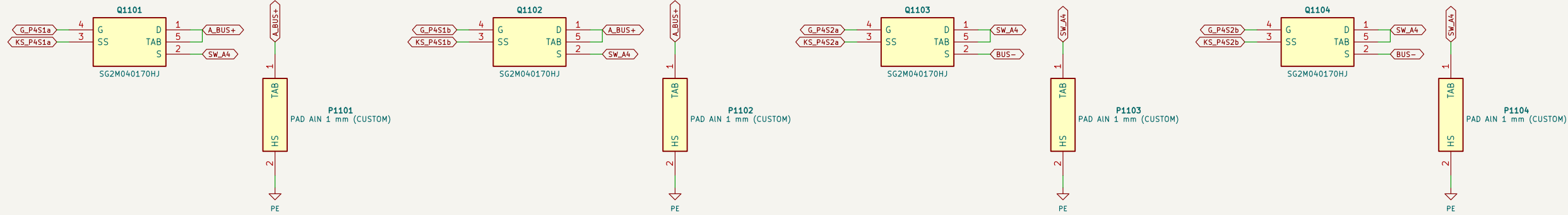


DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs - 3 x US1MH - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG A: LAYOUT RULES

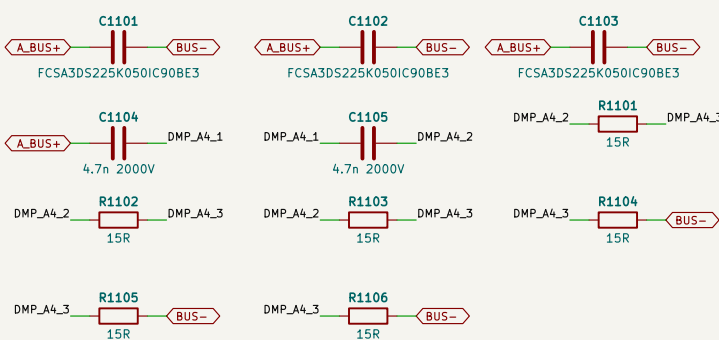
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (topside, gnd): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



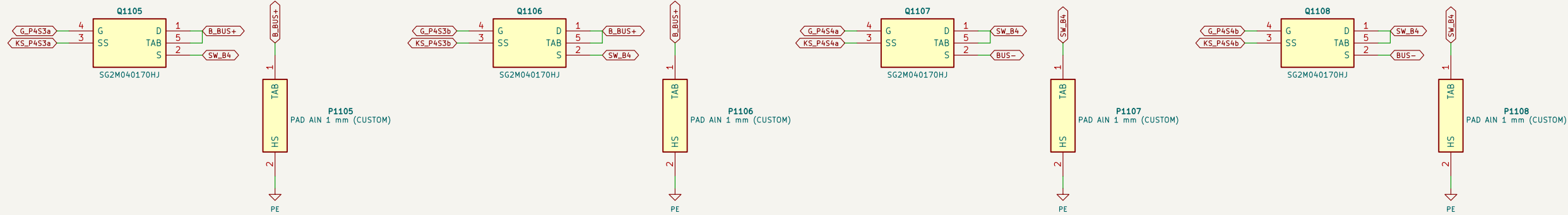
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



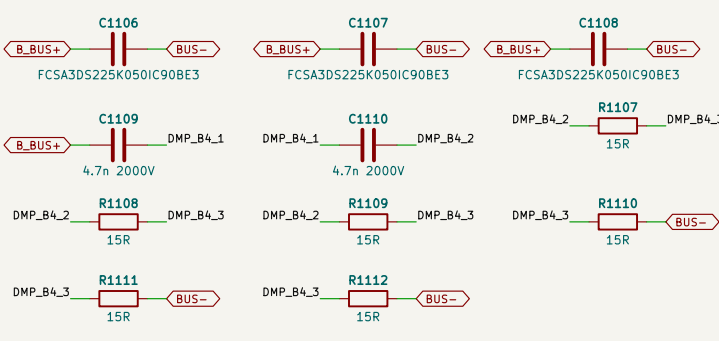
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series; loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



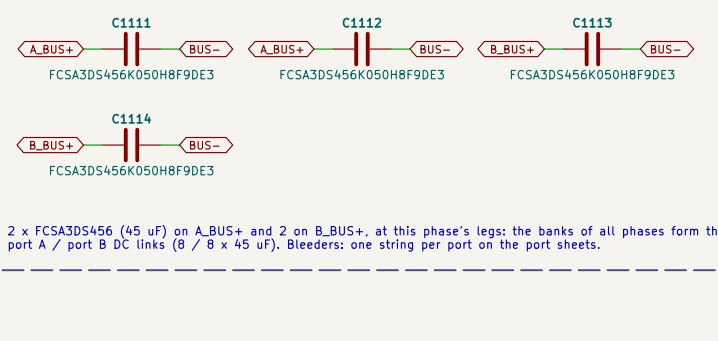
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



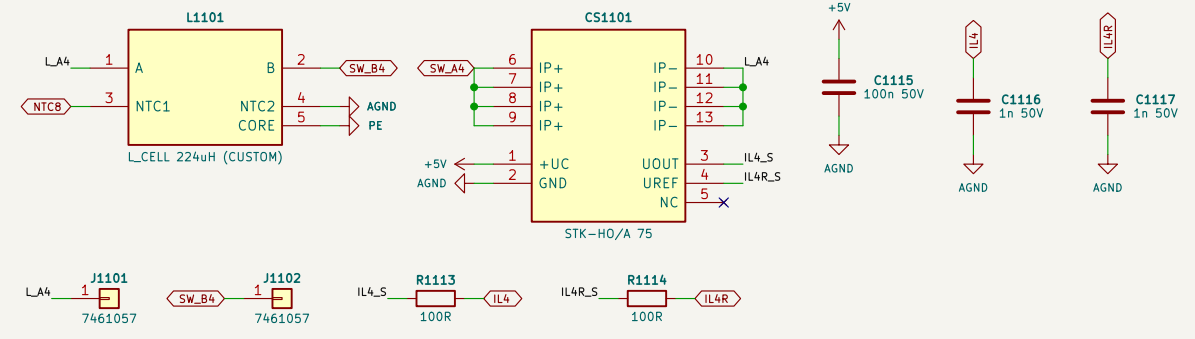
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series; loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



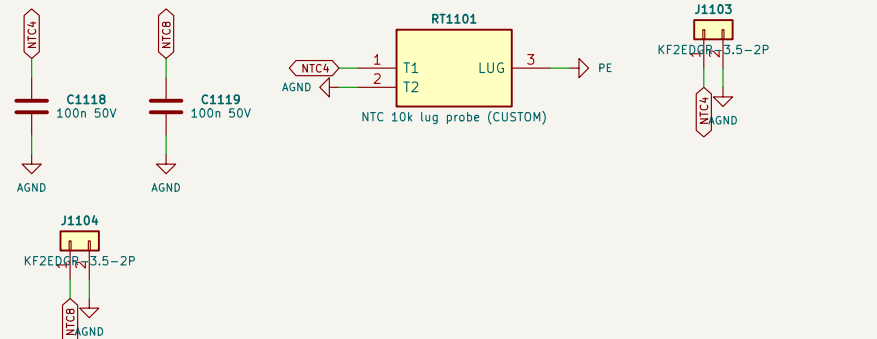
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



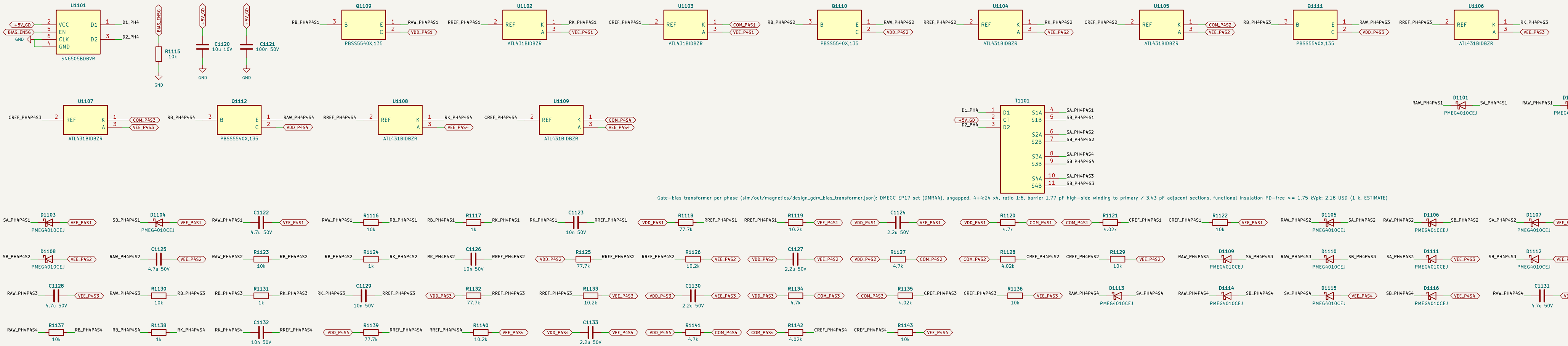
Lead A from SW.A4 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW.A -> SW.B) to stud
L.AA; lead B on stud SW.B4; IL4 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-0.5 V over +/-187.5 A;
100R = 1 nF at the source; IL4B = the sensor's own tref (pin 6); 100R = 1 nF; PVR-CTL references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

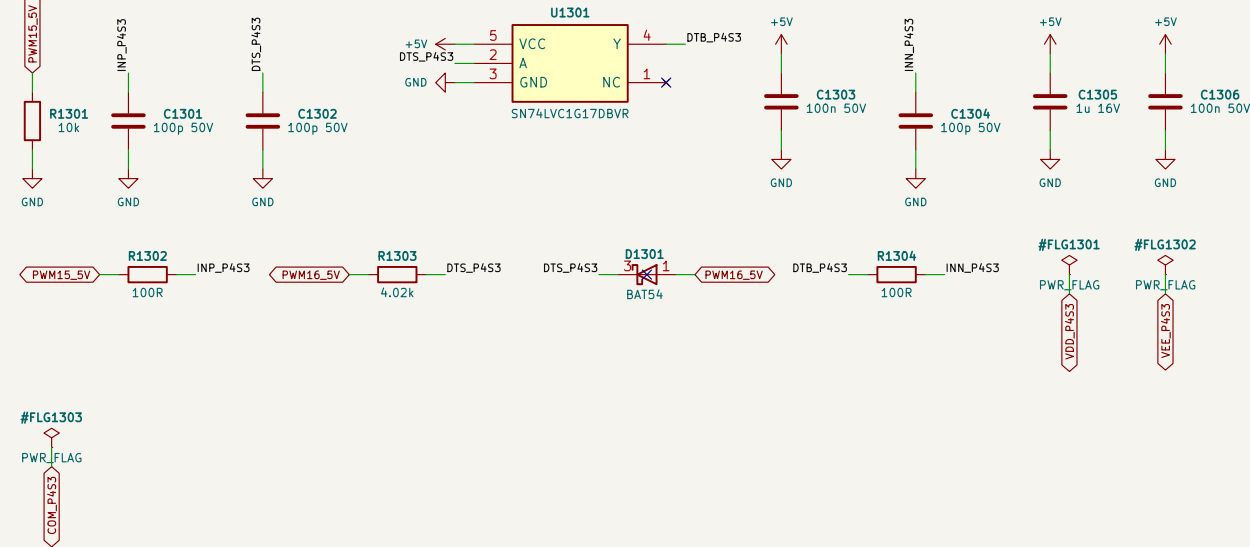


NTC4 = heatsink section of phase 4; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC6 = inductor winding NTC (functional to the winding); 100 nF at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)

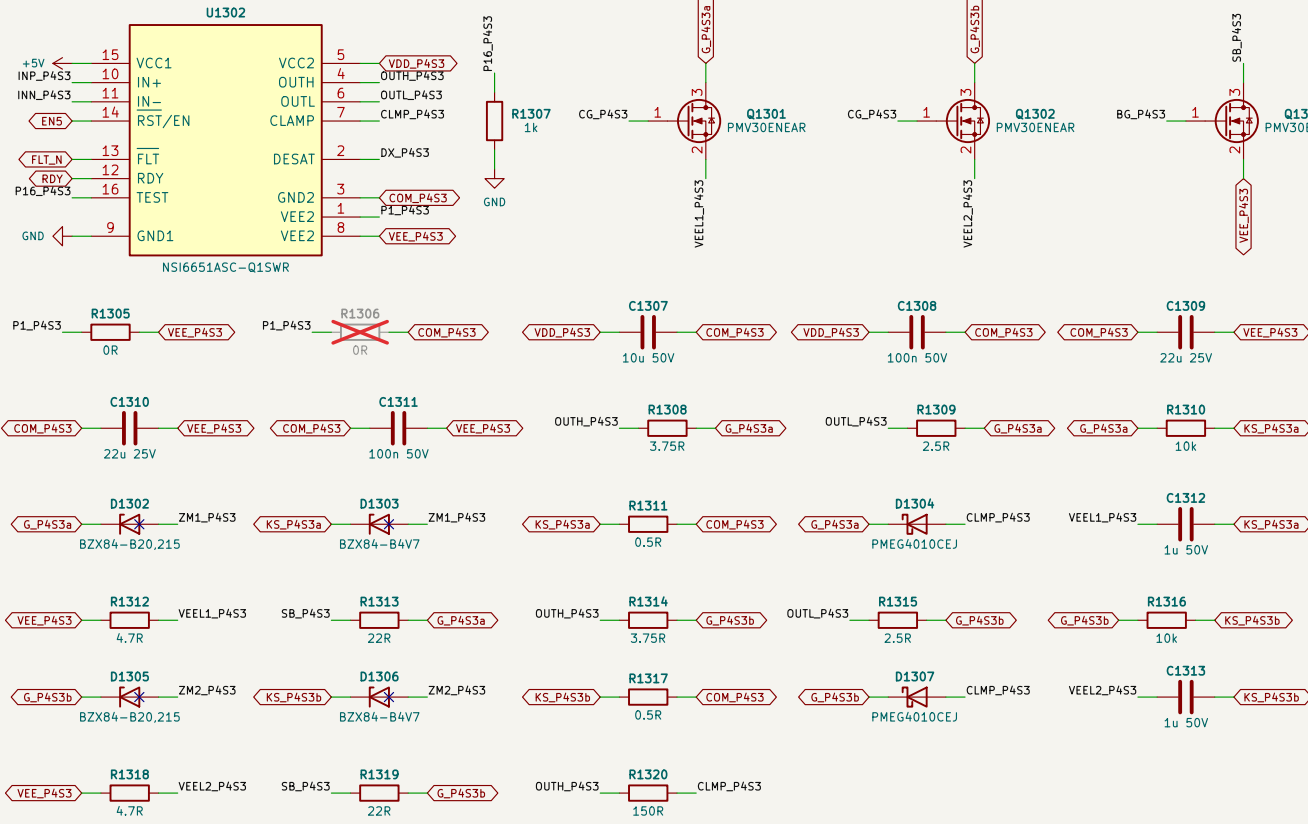


P4S3: LOGIC SIDE



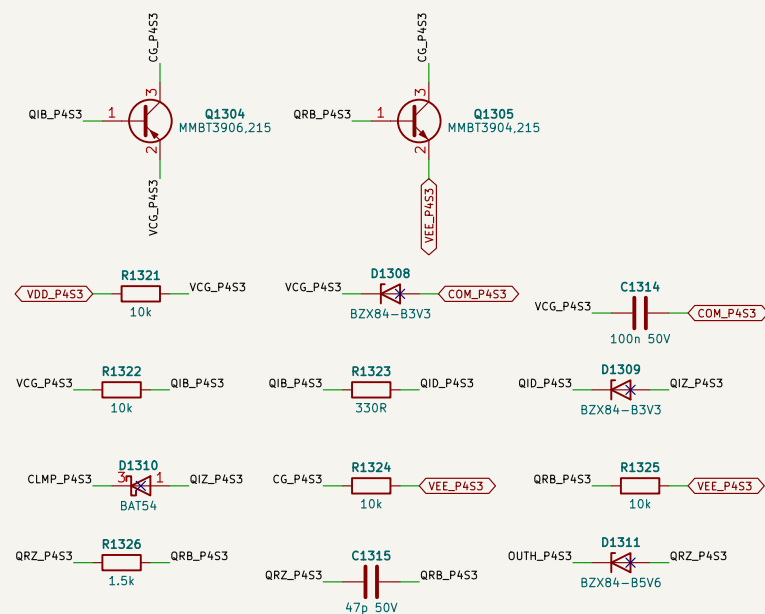
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P4S3: DRIVER AND 2 GATES



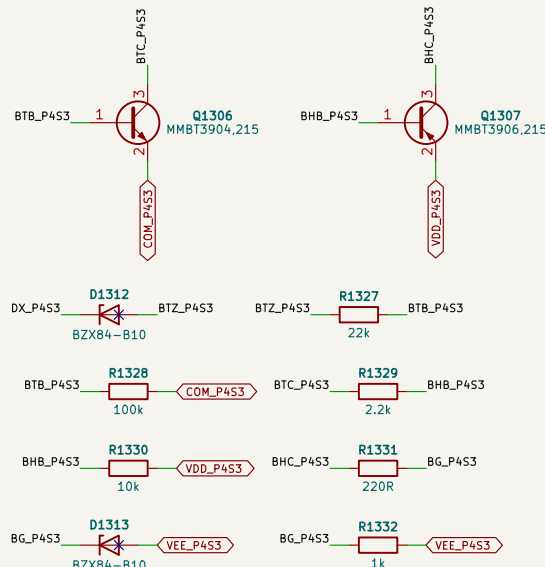
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P4S3: MILLER CLAMP CONTROL



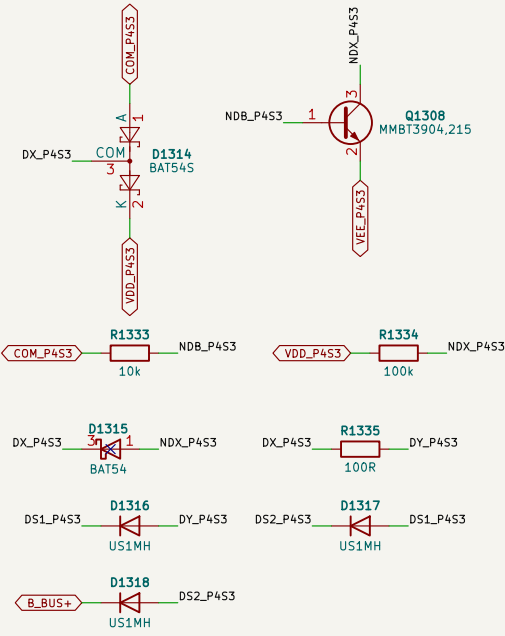
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P4S3: SHORT-CIRCUIT BOOSTER



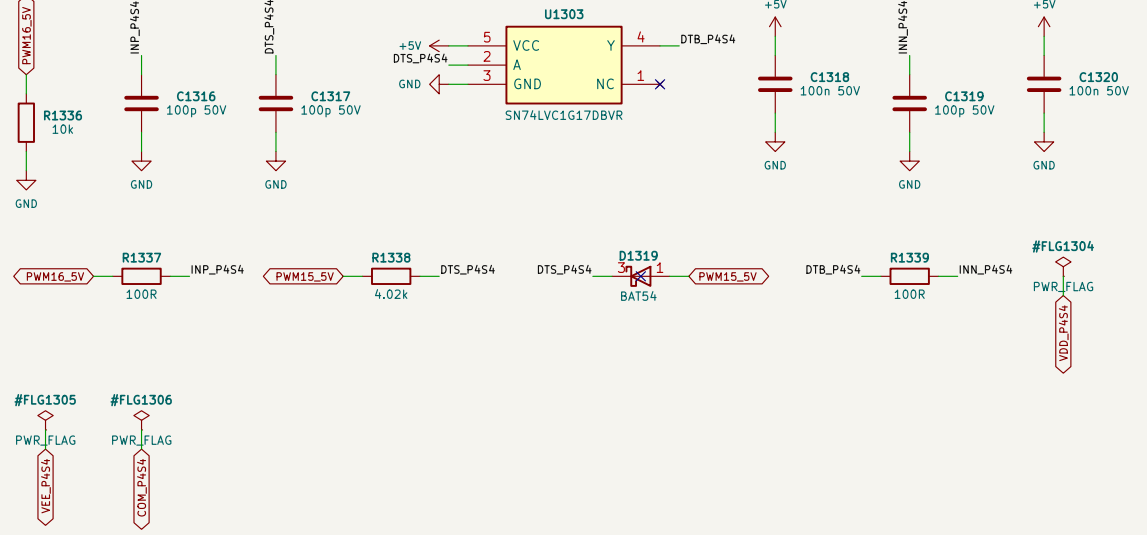
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P4S3: DESAT



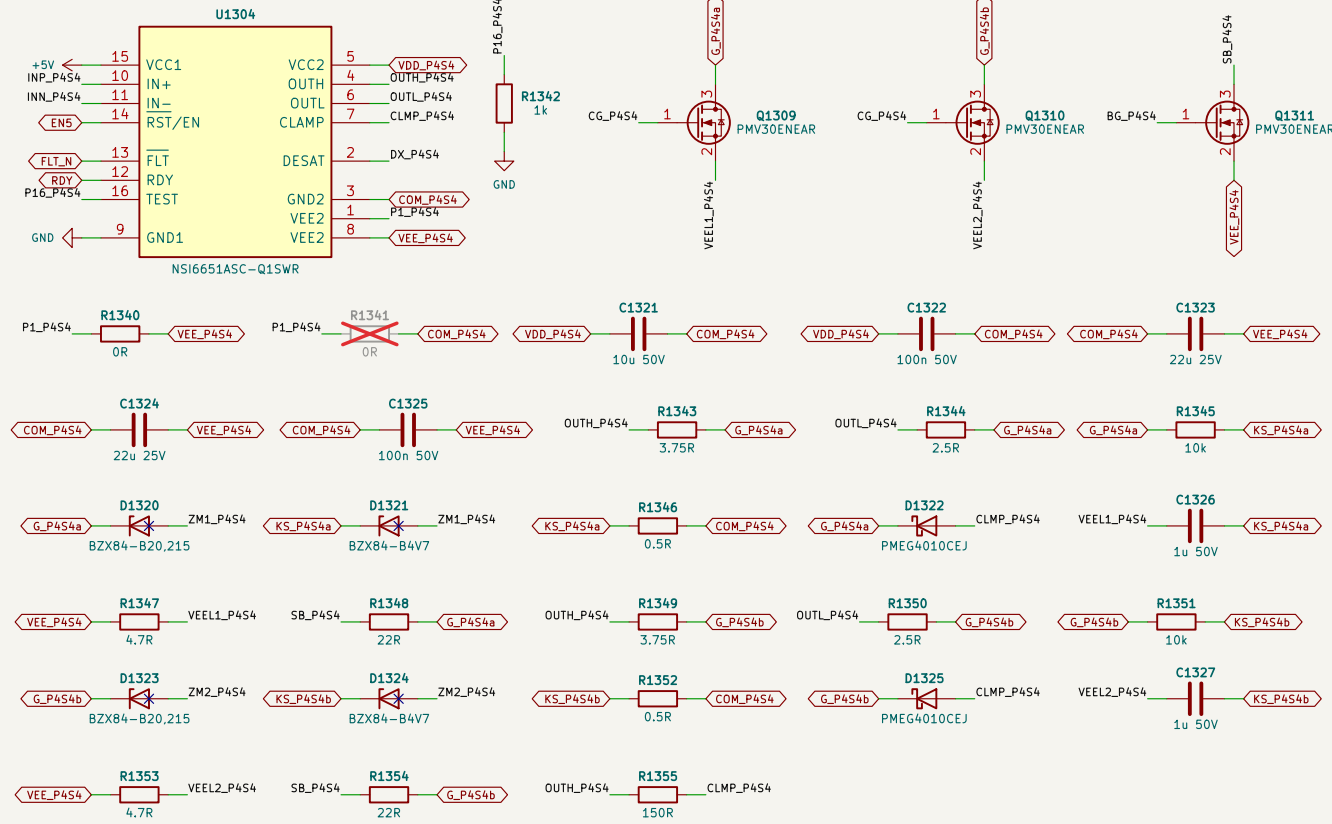
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; C0k footprint DNP.

P4S4: LOGIC SIDE



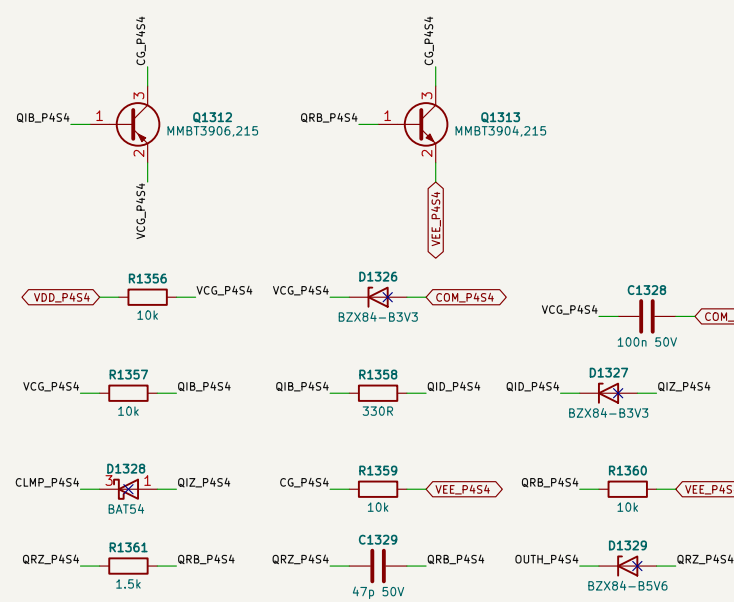
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P4S4: DRIVER AND 2 GATES



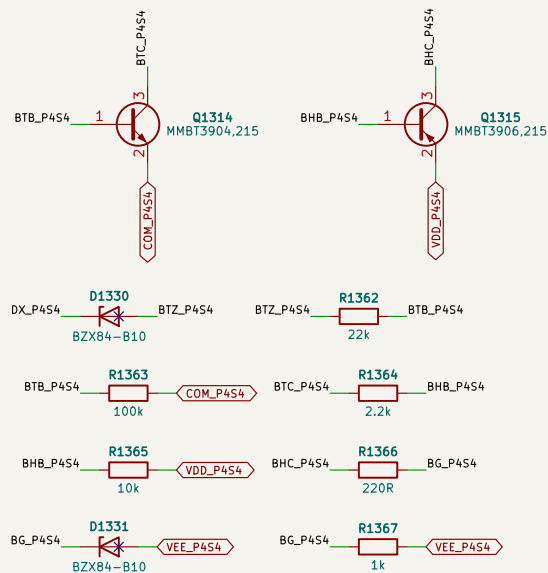
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P4S4: MILLER CLAMP CONTROL



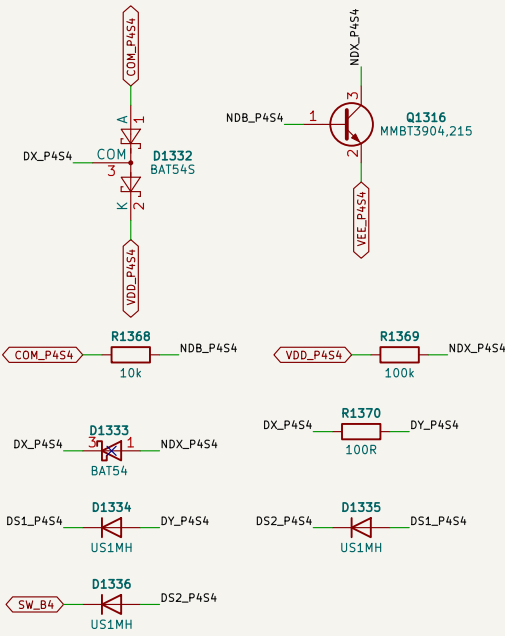
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P4S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P4S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; C0k footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (opposite, zero): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*;json; checks: design_check()
Sheet 14
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /13 Phase 4 leg B gate drive (S3, S4)/
File: 13_ph4-gdb.kicad_sch

Title: PV-PWR-4 power board - Phase 4 leg B gate drive (S3, S4)

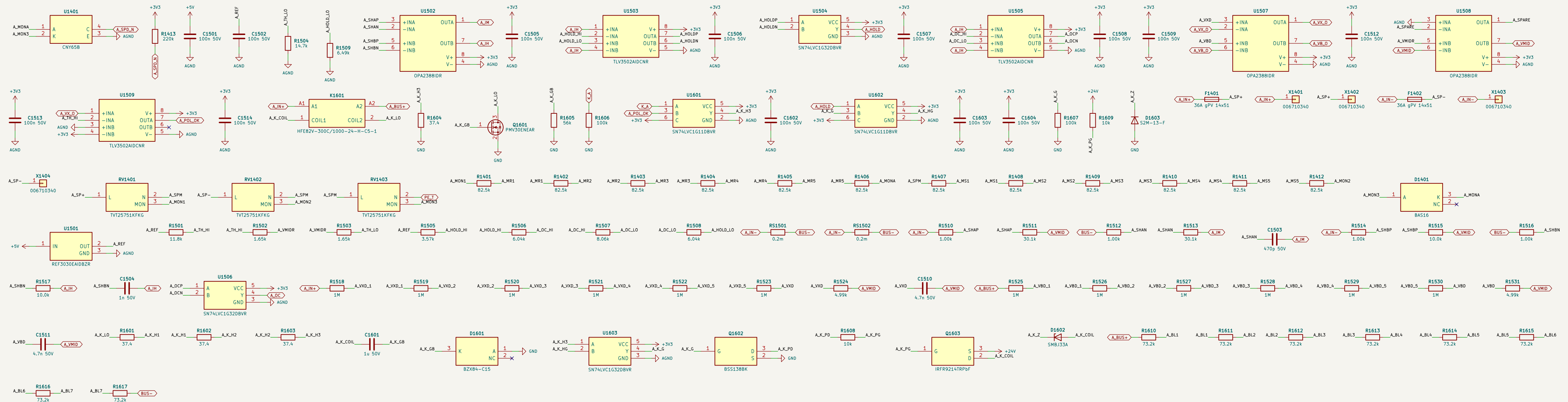
Size: A2 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 14/26

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{JIN+} / A_{JIN-}, rank A_{BUS+}. No fuse, no precharge: array I_{sc} ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA. Branch fuses ETI CH14x51 pV36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 15
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /14 Port A power path (pv)/
File: 14_leanA_power.kicad_sch

Title: PV-PWR-4 power board – Port A power path (pv)

Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 15/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Port A sensing and interlocks/
File: 15_leanA_ctrl.kicad_sch

Title: **PV-PWR-4 power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 16/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/
File: 16_leanA_drive.kicad_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 17/26

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

[illegible]

TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE-T, U_{eff} 3793 V pole-PE and pole-pole at 3 kA (calc. = 4.0 kV) => reinforced 6 kV / basic 4 kV, D=032; Imax 25 kA, Branch fuses E1 (sum) 45.5 gVp 36 A (30 kA / I₂ t₂ ms). Monitor loop through both fuses and all three links -> CNY658 -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 19
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 Port B sensing and interlocks/
File: 17_leanB_ctrl.kicad_sch

Title: **PV-PWR-4 power board – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

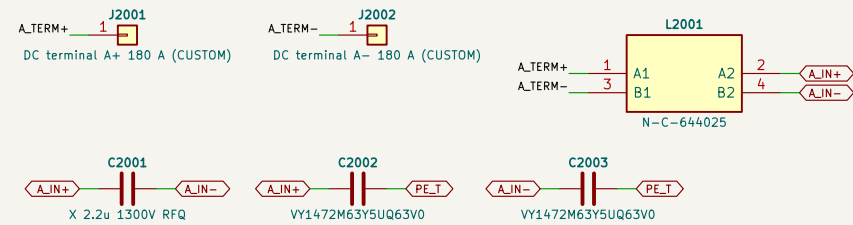
KiCad E.D.A. 10.0.6

Id: 19/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

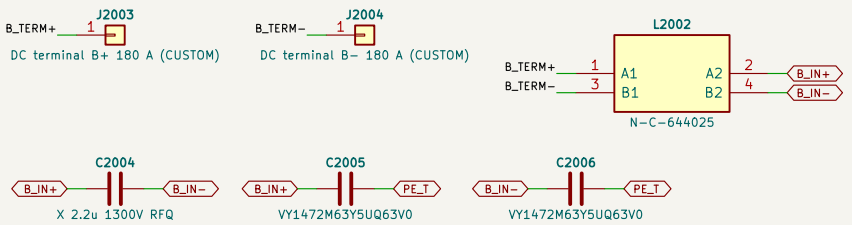
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 20 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /18 Port B coil drives/ File: 18_leanB_drive.kicad_sch		
Title: PV-PWR-4 power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 20/26	

PORT A TERMINALS AND EMI FILTER



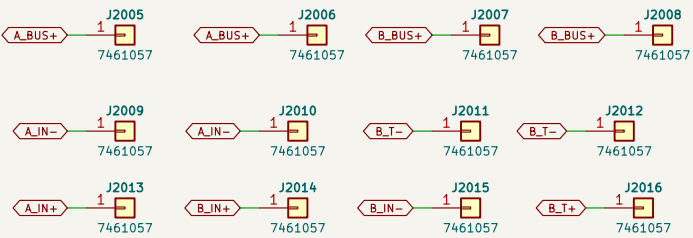
PV: chassis terminals A_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



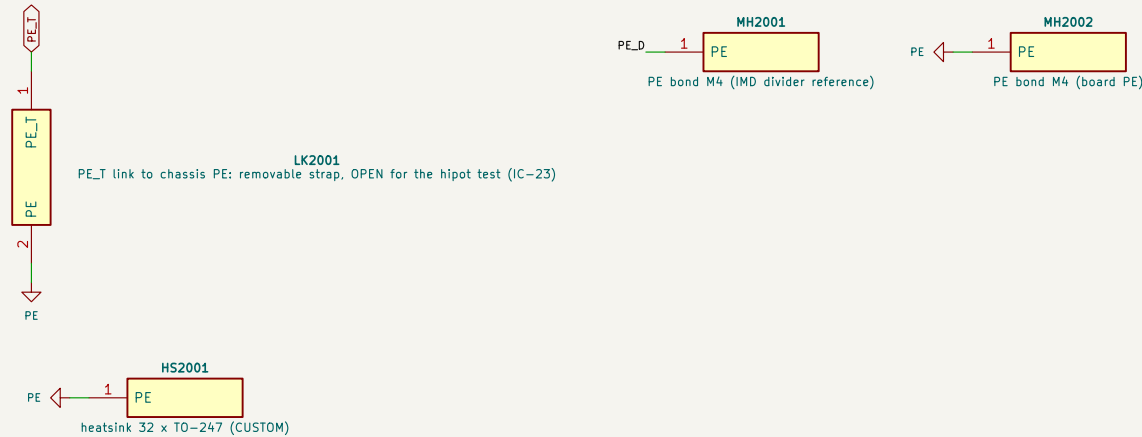
battery: chassis terminals B_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



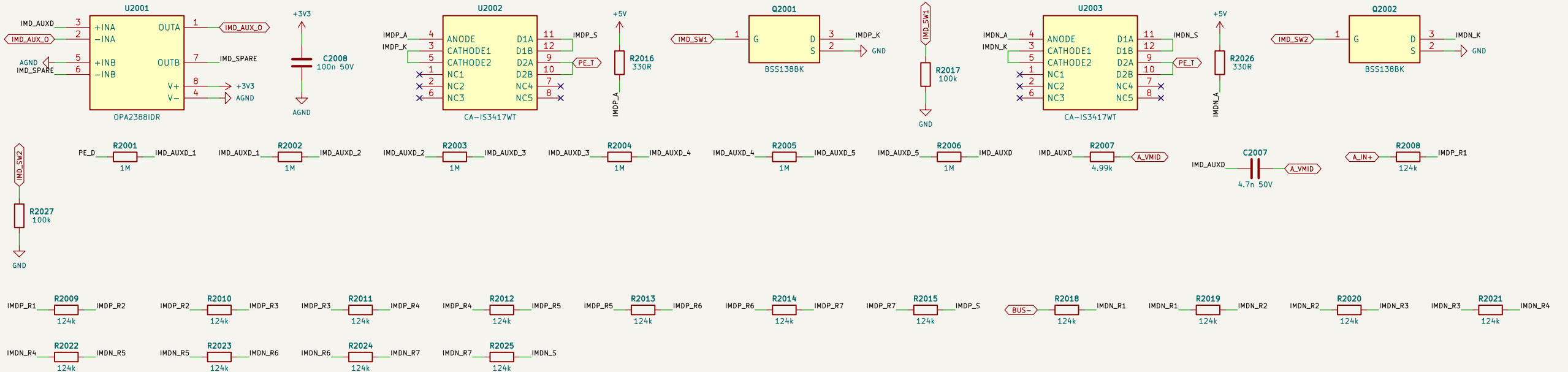
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V → +5V (LMR38020, 400 KHZ, 2 A)

U2101: LMR38020FDDAR

Pin connections:

- 3: VIN (24V)
- 2: EN (RT_5V)
- 4: RT/SYNC (RT_5V)
- 8: SW_5V
- 7: BOOT_5V
- 5: FB_5V
- 6: PG_5V
- 1: GND
- 9: EP

Components:

- C2101: 4.7uF 50V
- C2102: 4.7uF 50V
- C2103: 100nF 100V
- R2101: 64.9k
- L2101: 74437368150
- C2105: 22uF 16V
- C2106: 22uF 16V
- C2107: 22uF 16V
- R2102: 100k
- R2103: 24.9k
- C2104: 100nF 50V
- R2104: 100k
- TP2101: TP

NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % → 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)

gate bias of phases 3-4, ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)

gate bias of phases 3-4, ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP

The schematic shows the TPS62130RQTR buck converter circuit. The U2104 chip is connected as follows:

- Input:** +5V_GD (pin 11, 12, 10) connected to PVIN, PVIN, and AVIN.
- Control:** PG_SVG (pin 13) connected to EN; SS_3V3 (pin 9) connected to SS/TR; FSW (pin 7) connected to FSW; DEF (pin 8) connected to DEF.
- Output:** SW_3V3 (pin 1) connected to SW; FB_3V3 (pin 5) connected to FB; PG (pin 4) connected to PG.
- Grounds:** AGND (pin 6) connected to GND; PGND (pin 15, 16) connected to GND; EP (pin 17) connected to GND.

The output +3V3 is regulated by a feedback network consisting of R2111 (31.6k) and R2112 (10k) connected to pin 17 (EP). The circuit includes several capacitors: C2117 (10uF 25V) at the input, C2118 (100nF 50V) at the input, C2119 (470pF 50V) at the output, C2120 (22uF 10V) at the output, and C2121 (22uF 10V) at the output. The feedback network also includes L2103 (74438356022) and C2119 (470pF 50V).

EN = PG_SVG, FB $31.6k / 10k$ 0.1 % -> 3.240-3.433 V (≥ 3.20 V for the control board's reference). SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control ≤ 157 mA).

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP

The schematic shows the TPS62130RQTR regulator (U2104) configured to step down +5V_GD to +3V3. The input is connected to PVIN (pin 11), and the output is connected to SW_3V3 (pin 1). The feedback network consists of R2111 (31.6k) and R2112 (10k) connected to FB_3V3 (pin 14). The control logic is configured with EN (pin 13) to PG_SVG, SS/TR (pin 9) to SS_3V3, and FSW (pin 7) to GND. The output is filtered by C2120 (22uF 10V) and C2121 (22uF 10V). The input is filtered by C2117 (10uF 25V) and C2118 (100nF 50V). The output is also filtered by C2119 (470pF 50V). The ramp is controlled by L2103 (74438356022).

EN = PG_SVG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (≥ 3.20 V for the control board's reference). SS/TR 470p COG;
ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control ≤ 157 mA).

SELV 24 V TO THE CONTROL BOARD (J2)

AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3–29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

SELV 24 V TO THE CONTROL BOARD (J2)

AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3–29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basi							
Sheet: /19 Live 5 V / 3.3 V, SELV connector/ File: 19_supplies.kicad_sch							
Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector							
Size: A3		Date: 2026-10-05				Rev: A2	
KiCad E.D.A. 10.0.6						Id: 22/26	

+5V_GD SUPERVISOR ON RDY

The diagram illustrates the +5V_GD supervisor circuit on the RDY signal. The TPS3710DDCR (U2103) is configured with its VDD and SENSE pins connected to the +5V_GD supply. The OUT pin is connected to the RDY signal. The GND pins are connected to ground. A pull-up resistor R2109 (107k) is connected between the +5V_GD supply and the SUP5G_SNS pin. The R2110 (10k) resistor is connected between the SUP5G_SNS pin and ground. The C2115 (100n 50V) capacitor is connected between the +5V_GD supply and ground. The C2116 (1n 50V) capacitor is connected between the SUP5G_SNS pin and ground.

The diagrams illustrate the +3V3 discharge path during a rail failure. The top row shows the normal state with +3V3 and +24V rails. The bottom row shows the failure state where PG_5V0 is low, disabling the TPS62130 and allowing the 100k resistor to pull the +3V3 rail down to GND through the B2R01 diode.

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)

The diagram illustrates the discharge path for the +3V3 rail when the rails fail. The circuit includes a 1.2R resistor (R2113) and a 100k resistor (R2114) connected to the +3V3 and +24V rails, respectively. The discharge path is controlled by the PG_5V0 signal, which is connected to the gate of the MOSFET (Q2102, 2N7002BK). The MOSFET is also connected to the +3V3 rail. The discharge path is shown in two states: normal operation (top) and failure state (bottom). In the normal state, the +3V3 rail is supplied by the 1.2R resistor, and the +24V rail is supplied by the 100k resistor. In the failure state, the PG_5V0 signal is low, disabling the MOSFET, and the +24V rail is pulled down to GND by the 100k resistor. The +3V3 rail is pulled down to GND by the 1.2R resistor and the 100k resistor.

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 22		
Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basis		
Sheet: /19 Live 5 V / 3.3 V, SELV connector/ File: 19_supplies.kicad_sch		
Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 22/26	

TAP A_IN+



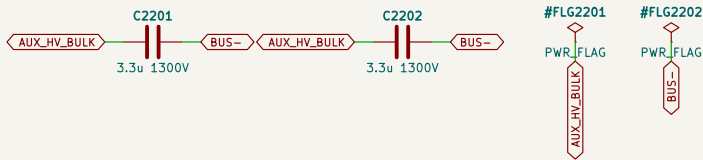
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

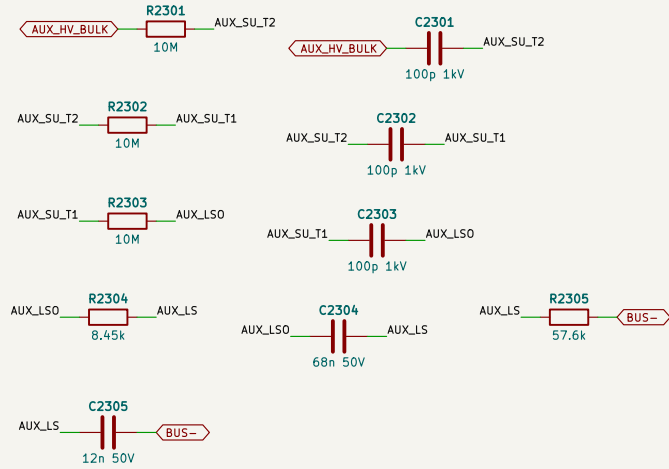
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /20 AUX75 input taps, OR, film/
File: 20_aux75_input.kicad_sch

Title: PV-PWR-4 power board – AUX75 input taps, OR, film

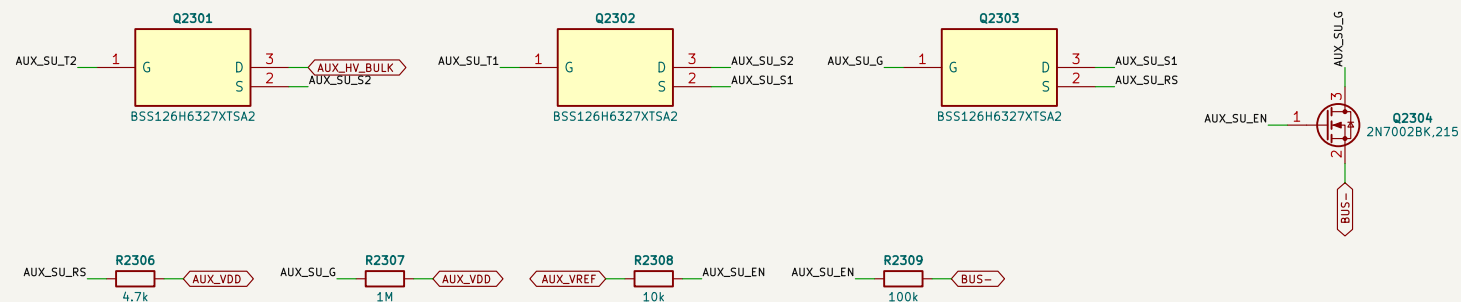
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 23/26

COMPENSATED TAP STRING



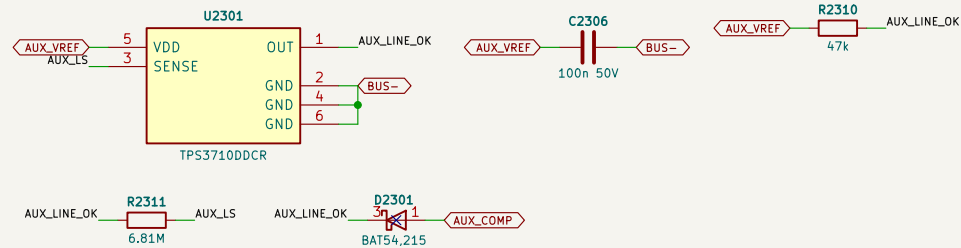
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



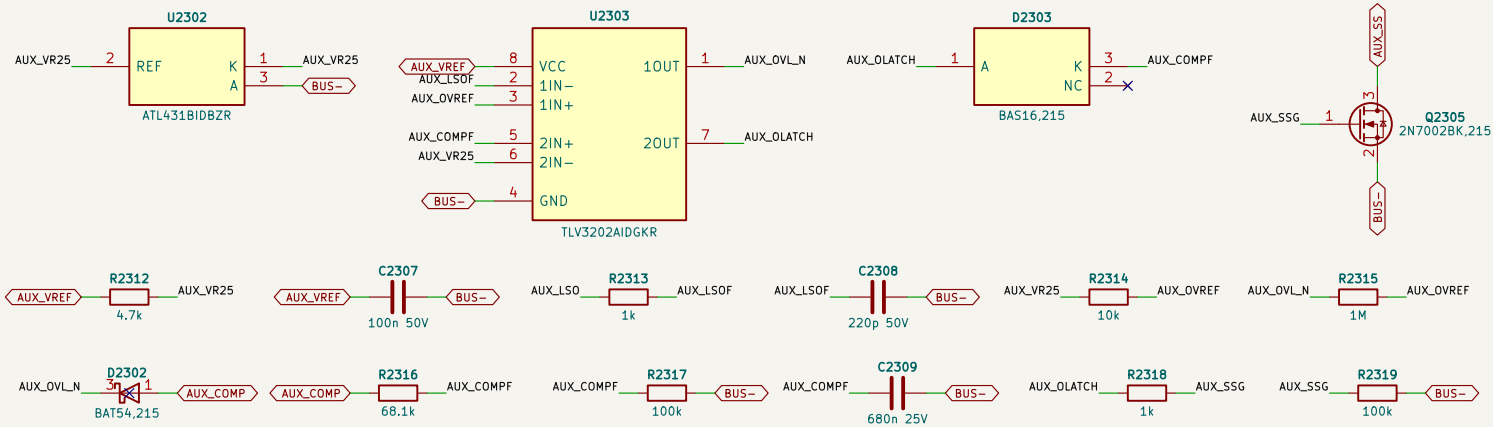
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



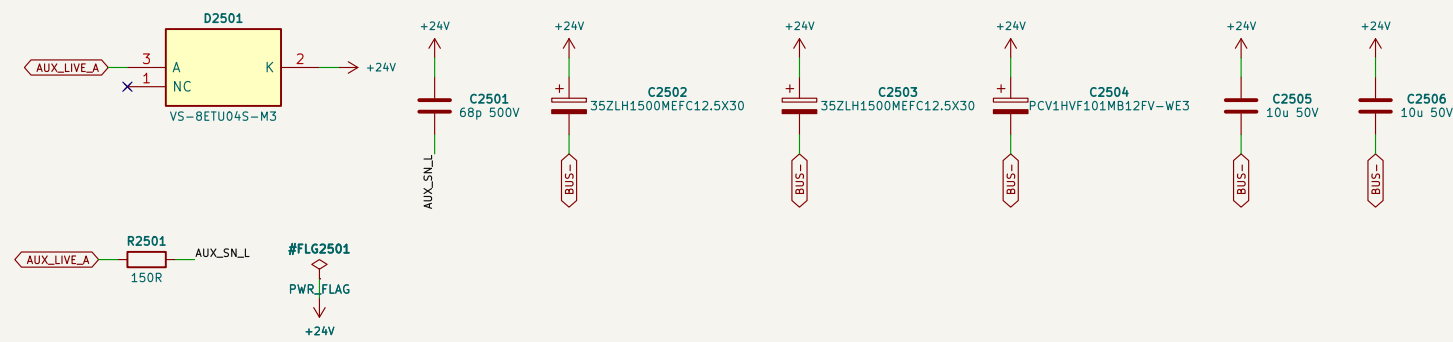
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



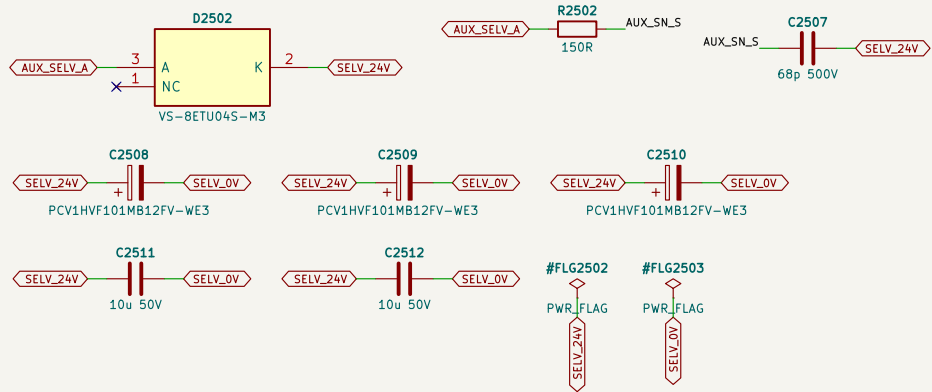
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

LIVE 24 V RECTIFIER, HOLD-UP



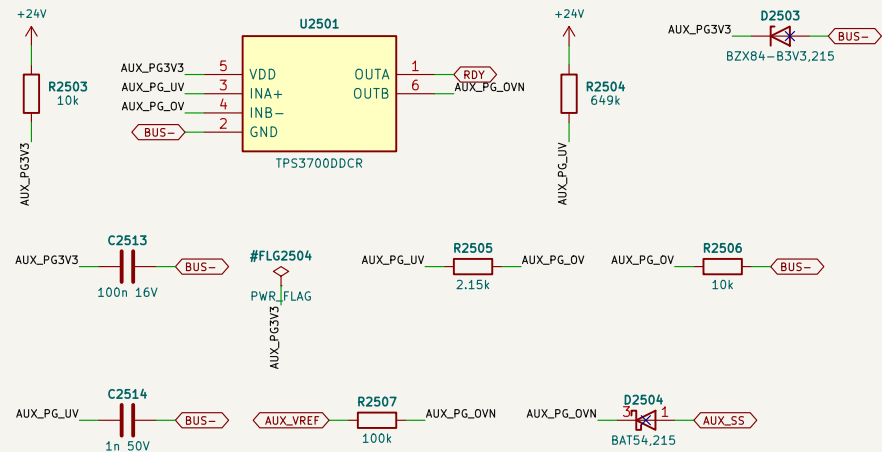
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)